1 Univariate analysis

Since high school, young people are taught the importance of measuring and describing variables as elementary as height, weight, income distribution or student achievement. **This are quantitative data analysis.**

Analyzes of this type can be **univariate**, that is, assess a single variable, **bivariate** when we focus on finding correlations between two variables, or **multivariate**, which consists of studying more than two variables.

When we analyze a variable and use **descriptive statistics** to calculate **measures of central tendency** such as a mean, a median, and a mode; Likewise, when we estimate **the dispersion or variability** as variance or standard deviation, we use a **univariate analysis.**

1.1 What are dependent variables and independent variables?

In an investigation, the **independent variable** is a "controlled and known" variable that we will manipulate in our study.

For example, measuring the dose of a drug that is believed to help diabetics regulate blood sugar.

The **dependent variable**, on the other hand, is the result that will be measured as the degree of sugar in the blood that will be obtained after administering a medication.

If we want, for example, to know the effects of carbohydrate consumption on the weight of young people, our "known" or **independent variable** is the amount of carbohydrates that we will supply in grams. The “result” or **dependent variable** will be the weight of the boys.

1.2 What is univariate analysis?

A univariate study is the simplest way to analyze data.

We perform a univariate study when we use **descriptive statistics** to show the behavior of the variable.

We can thus analyze the average height of a group of students, their weight or student performance. We generally emphasize one variable at a time.

1.3 Descriptive statistics in categorical data

1.3.1 Frequency analysis

This involves calculating the frequency and proportion of each category in the dataset. It can be done using basic summary statistics such as counts and percentages.

In [1]:
```
library(gapminder)
library(dplyr)
library(ggplot2)
```

```
Warning message:
"package 'gapminder' was built under R version 3.6.3"
Warning message:
"package 'dplyr' was built under R version 3.6.3"

Attaching package: 'dplyr'


The following objects are masked from 'package:stats':

    filter, lag


The following objects are masked from 'package:base':

    intersect, setdiff, setequal, union


Warning message:
"package 'ggplot2' was built under R version 3.6.3"
```

In [2]:
```
gapminder
```

A tibble: 1704 × 6

country	continent	year	lifeExp	pop	gdpPercap
<fct>	<fct>	<int>	<dbl>	<int>	<dbl>
Afghanistan	Asia	1952	28.801	8425333	779.4453
Afghanistan	Asia	1957	30.332	9240934	820.8530
Afghanistan	Asia	1962	31.997	10267083	853.1007
Afghanistan	Asia	1967	34.020	11537966	836.1971
Afghanistan	Asia	1972	36.088	13079460	739.9811
Afghanistan	Asia	1977	38.438	14880372	786.1134
Afghanistan	Asia	1982	39.854	12881816	978.0114
Afghanistan	Asia	1987	40.822	13867957	852.3959
Afghanistan	Asia	1992	41.674	16317921	649.3414
Afghanistan	Asia	1997	41.763	22227415	635.3414
Afghanistan	Asia	2002	42.129	25268405	726.7341
Afghanistan	Asia	2007	43.828	31889923	974.5803
Albania	Europe	1952	55.230	1282697	1601.0561
Albania	Europe	1957	59.280	1476505	1942.2842
Albania	Europe	1962	64.820	1728137	2312.8890
Albania	Europe	1967	66.220	1984060	2760.1969
Albania	Europe	1972	67.690	2263554	3313.4222
Albania	Europe	1977	68.930	2509048	3533.0039
Albania	Europe	1982	70.420	2780097	3630.8807
Albania	Europe	1987	72.000	3075321	3738.9327
Albania	Europe	1992	71.581	3326498	2497.4379
Albania	Europe	1997	72.950	3428038	3193.0546
Albania	Europe	2002	75.651	3508512	4604.2117
Albania	Europe	2007	76.423	3600523	5937.0295
Algeria	Africa	1952	43.077	9279525	2449.0082
Algeria	Africa	1957	45.685	10270856	3013.9760
Algeria	Africa	1962	48.303	11000948	2550.8169
Algeria	Africa	1967	51.407	12760499	3246.9918
Algeria	Africa	1972	54.518	14760787	4182.6638
Algeria	Africa	1977	58.014	17152804	4910.4168
⋮	⋮	⋮	⋮	⋮	⋮
Yemen, Rep.	Asia	1982	49.113	9657618	1977.5570
Yemen, Rep.	Asia	1987	52.922	11219340	1971.7415
Yemen, Rep.	Asia	1992	55.599	13367997	1879.4967

country	continent	year	lifeExp	pop	gdpPercap
<fct>	<fct>	<int>	<dbl>	<int>	<dbl>
Yemen, Rep.	Asia	1997	58.020	15826497	2117.4845
Yemen, Rep.	Asia	2002	60.308	18701257	2234.8208
Yemen, Rep.	Asia	2007	62.698	22211743	2280.7699
Zambia	Africa	1952	42.038	2672000	1147.3888
Zambia	Africa	1957	44.077	3016000	1311.9568
Zambia	Africa	1962	46.023	3421000	1452.7258
Zambia	Africa	1967	47.768	3900000	1777.0773
Zambia	Africa	1972	50.107	4506497	1773.4983
Zambia	Africa	1977	51.386	5216550	1588.6883
Zambia	Africa	1982	51.821	6100407	1408.6786
Zambia	Africa	1987	50.821	7272406	1213.3151
Zambia	Africa	1992	46.100	8381163	1210.8846
Zambia	Africa	1997	40.238	9417789	1071.3538
Zambia	Africa	2002	39.193	10595811	1071.6139
Zambia	Africa	2007	42.384	11746035	1271.2116
Zimbabwe	Africa	1952	48.451	3080907	406.8841
Zimbabwe	Africa	1957	50.469	3646340	518.7643
Zimbabwe	Africa	1962	52.358	4277736	527.2722
Zimbabwe	Africa	1967	53.995	4995432	569.7951
Zimbabwe	Africa	1972	55.635	5861135	799.3622
Zimbabwe	Africa	1977	57.674	6642107	685.5877
Zimbabwe	Africa	1982	60.363	7636524	788.8550
Zimbabwe	Africa	1987	62.351	9216418	706.1573
Zimbabwe	Africa	1992	60.377	10704340	693.4208
Zimbabwe	Africa	1997	46.809	11404948	792.4500
Zimbabwe	Africa	2002	39.989	11926563	672.0386
Zimbabwe	Africa	2007	43.487	12311143	469.7093

We can explore categorical variables in the data using several functions of the dplyr package. For example, we can count the unique values of a categorical variable and sort them by relative frequency as follows:

In [3]:

```
gapminder %>%
  count(continent, year, sort = TRUE) %>%
  mutate(prop = n / sum(n))
```

A tibble: 60 × 4

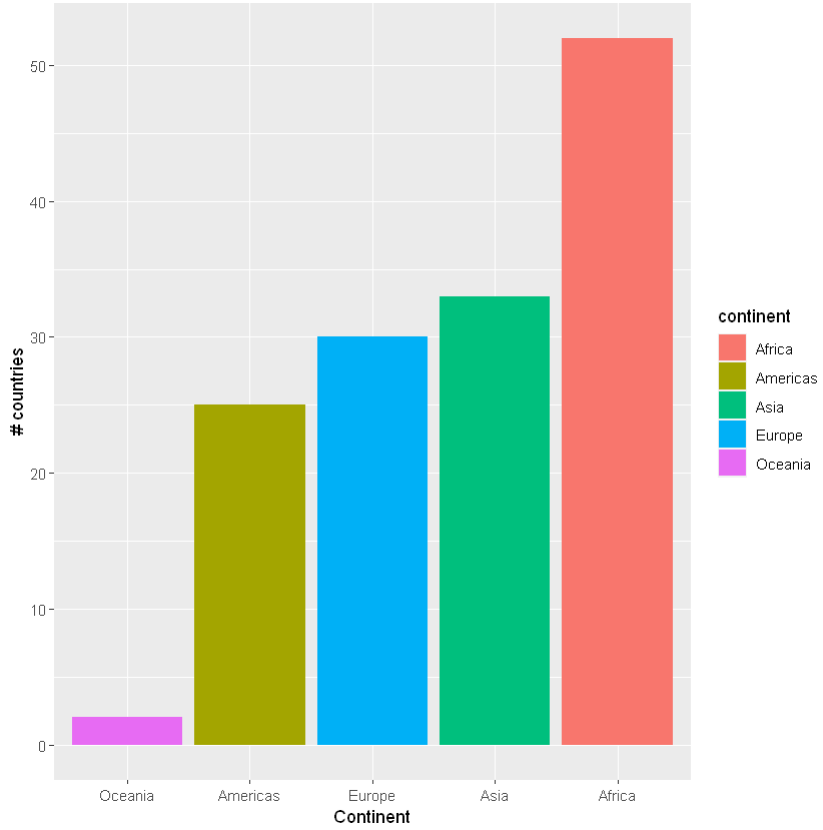
continent	year	n	prop
<fct>	<int>	<int>	<dbl>
Africa	1952	52	0.030516432
Africa	1957	52	0.030516432
Africa	1962	52	0.030516432
Africa	1967	52	0.030516432
Africa	1972	52	0.030516432
Africa	1977	52	0.030516432
Africa	1982	52	0.030516432
Africa	1987	52	0.030516432
Africa	1992	52	0.030516432
Africa	1997	52	0.030516432
Africa	2002	52	0.030516432
Africa	2007	52	0.030516432
Asia	1952	33	0.019366197
Asia	1957	33	0.019366197
Asia	1962	33	0.019366197
Asia	1967	33	0.019366197
Asia	1972	33	0.019366197
Asia	1977	33	0.019366197
Asia	1982	33	0.019366197
Asia	1987	33	0.019366197
Asia	1992	33	0.019366197
Asia	1997	33	0.019366197
Asia	2002	33	0.019366197
Asia	2007	33	0.019366197
Europe	1952	30	0.017605634
Europe	1957	30	0.017605634
Europe	1962	30	0.017605634
Europe	1967	30	0.017605634
Europe	1972	30	0.017605634
Europe	1977	30	0.017605634
Europe	1982	30	0.017605634
Europe	1987	30	0.017605634
Europe	1992	30	0.017605634
Europe	1997	30	0.017605634

continent	year	n	prop
<fct>	<int>	<int>	<dbl>
Europe	2002	30	0.017605634
Europe	2007	30	0.017605634
Americas	1952	25	0.014671362
Americas	1957	25	0.014671362
Americas	1962	25	0.014671362
Americas	1967	25	0.014671362
Americas	1972	25	0.014671362
Americas	1977	25	0.014671362
Americas	1982	25	0.014671362
Americas	1987	25	0.014671362
Americas	1992	25	0.014671362
Americas	1997	25	0.014671362
Americas	2002	25	0.014671362
Americas	2007	25	0.014671362
Oceania	1952	2	0.001173709
Oceania	1957	2	0.001173709
Oceania	1962	2	0.001173709
Oceania	1967	2	0.001173709
Oceania	1972	2	0.001173709
Oceania	1977	2	0.001173709
Oceania	1982	2	0.001173709
Oceania	1987	2	0.001173709
Oceania	1992	2	0.001173709
Oceania	1997	2	0.001173709
Oceania	2002	2	0.001173709
Oceania	2007	2	0.001173709

This will give us a table that counts the number of countries in each continent and their relative frequency. We can visualize this information with a bar chart using ggplot2:

In [4]:
```
library(ggplot2)

gapminder %>%
  filter(year == 2007) %>%
  count(continent) %>%
  ggplot(aes(x = reorder(continent, n), y = n, fill = continent)) +
  geom_col() +
  labs(x = "Continent", y = "# countries")
```



We can analyze categorical variables using dplyr's group_by() function. For example, we can group the data by continent and calculate the mean life expectancy using the following function:

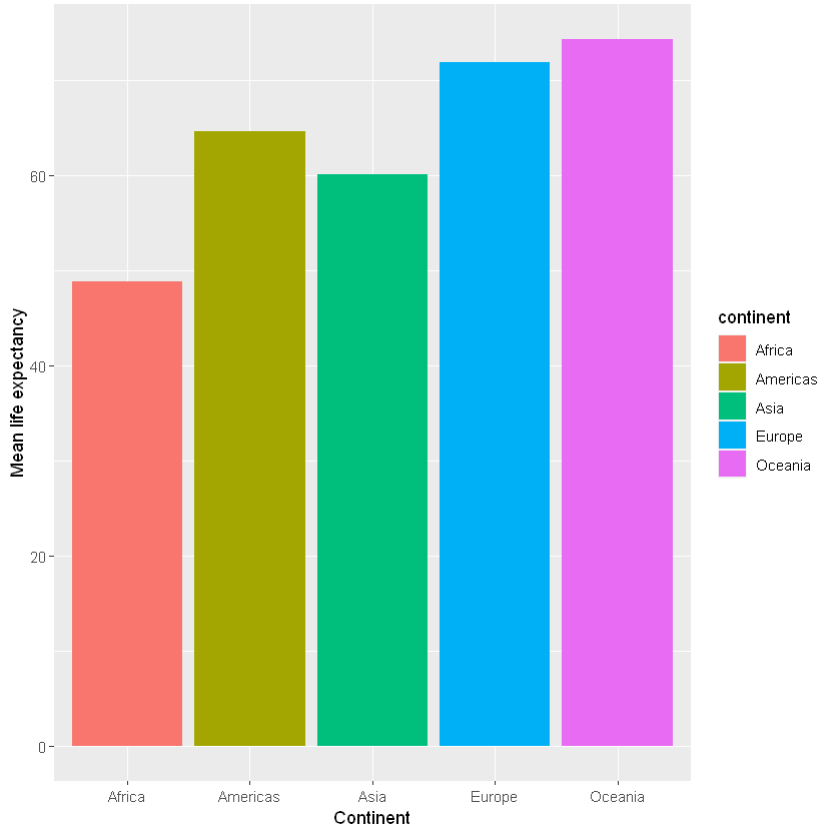
In [5]:
```
gapminder %>%
  group_by(continent) %>%
  summarise(mean_lifeExp = mean(lifeExp, na.rm = TRUE))
```

A tibble: 5 × 2

continent	mean_lifeExp
<fct>	<dbl>
Africa	48.86533
Americas	64.65874
Asia	60.06490
Europe	71.90369
Oceania	74.32621

This will give us a table showing the average life expectancy for each continent. We can visualize this information with a bar chart using ggplot2:

In [6]:
```
gapminder %>%
  group_by(continent) %>%
  summarise(mean_lifeExp = mean(lifeExp, na.rm = TRUE)) %>%
  ggplot(aes(x = continent, y = mean_lifeExp, fill = continent)) +
  geom_col() +
  labs(x = "Continent", y = "Mean life expectancy")
```



1.3.2 Cross-tabulation:

This involves examining the relationship between two or more categorical variables by creating a contingency table that shows the frequency of each combination of categories. Cross-tabulation can help identify patterns and relationships between variables.

Suppose we want to explore the relationship between the continent and year variables in the gapminder dataset. We can create a contingency table that shows the frequency of each combination of categories using the table() function:

In [7]:
```
table(gapminder$continent, gapminder$year)

#or

ct <- xtabs(~ continent + year, data = gapminder)
ct
```

```
          1952 1957 1962 1967 1972 1977 1982 1987 1992 1997 2002 2007
  Africa    52   52   52   52   52   52   52   52   52   52   52   52
  Americas  25   25   25   25   25   25   25   25   25   25   25   25
  Asia      33   33   33   33   33   33   33   33   33   33   33   33
  Europe    30   30   30   30   30   30   30   30   30   30   30   30
  Oceania    2    2    2    2    2    2    2    2    2    2    2    2

          year
continent  1952 1957 1962 1967 1972 1977 1982 1987 1992 1997 2002 2007
  Africa    52   52   52   52   52   52   52   52   52   52   52   52
  Americas  25   25   25   25   25   25   25   25   25   25   25   25
  Asia      33   33   33   33   33   33   33   33   33   33   33   33
  Europe    30   30   30   30   30   30   30   30   30   30   30   30
  Oceania    2    2    2    2    2    2    2    2    2    2    2    2
```

In [8]:
```
library(ggplot2)
ct <- xtabs(~ continent + year, data = gapminder)

ct

as.data.frame(ct)

ggplot(data = as.data.frame(ct), aes(x = year, y = continent, fill = Freq)) +
  geom_tile() +
  scale_fill_gradient(low = "#FEE08B", high = "#A50026") +
  labels(x = "Year", y = "Continent", fill = "Frequency")
```

Americas	1957	25
Asia	1957	33
Europe	1957	30
Oceania	1957	2
Africa	1962	52
Americas	1962	25
Asia	1962	33
Europe	1962	30
Oceania	1962	2
Africa	1967	52
Americas	1967	25
Asia	1967	33
Europe	1967	30

A better example would be:

In [9]:

```
library(ggplot2)

# Set seed for reproducibility
set.seed(616145)

# Generate random data for the dataset
n <- 1000
sex <- sample(c("Male", "Female"), n, replace = TRUE)
smoker <- sample(c("Yes", "No"), n, replace = TRUE)
data <- data.frame(sex = sex, smoker = smoker)

data
```

A data.frame: 1000 × 2

sex	smoker
<fct>	<fct>
Female	No
Male	Yes
Female	No
Female	Yes
Male	Yes
Female	Yes
Male	Yes
Male	No
Male	No
Female	No
Male	No
Male	No
Male	No
Male	No
Female	No
Female	Yes
Male	Yes
Female	Yes
Male	Yes
Male	Yes
Male	Yes
Male	No
Female	Yes
Female	Yes
Female	No
Male	No
Male	No
Female	No
Female	No
Male	No
⋮	⋮
Male	No
Female	Yes

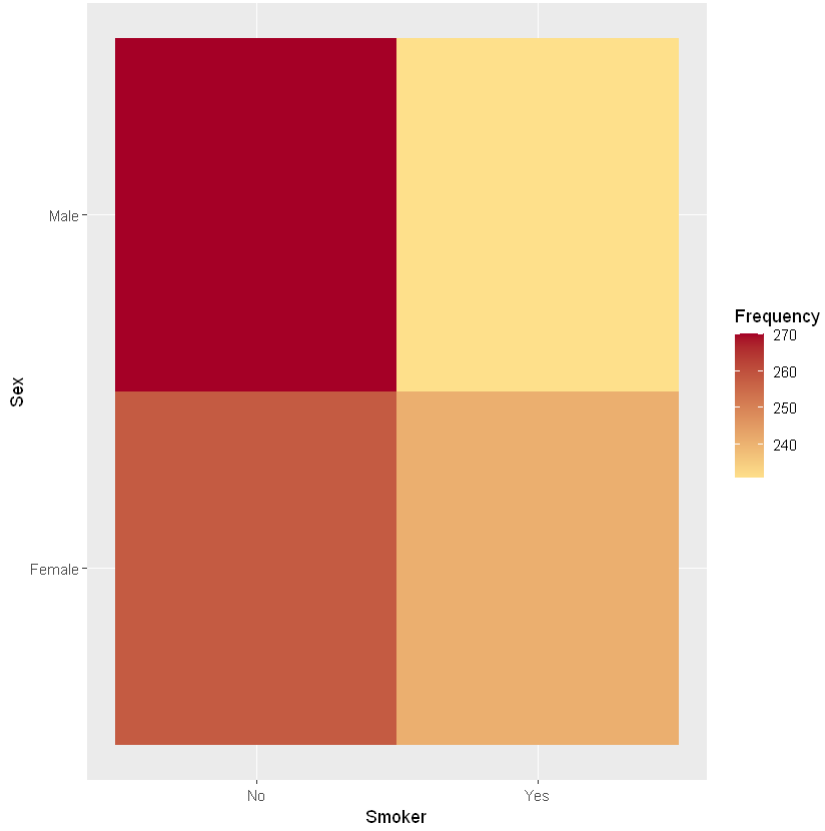
sex	smoker
<fct>	<fct>
Male	No
Female	Yes
Male	Yes
Female	No
Male	No
Female	Yes
Male	No
Female	Yes
Male	Yes
Female	Yes
Male	Yes
Female	Yes
Female	No
Male	No
Female	Yes
Male	No
Male	Yes
Female	Yes
Female	No
Male	No
Male	Yes
Female	No
Male	No
Male	Yes
Female	No
Male	No
Male	No
Female	No

In [10]:
```
# Create contingency table
ct <- xtabs(~ sex + smoker, data = data)

ct

# Create heatmap with ggplot2
ggplot(data = as.data.frame(ct), aes(x = smoker, y = sex, fill = Freq)) +
  geom_tile() +
  scale_fill_gradient(low = "#FEE08B", high = "#A50026") +
  labels(x = "Smoker", y = "Sex", fill = "Frequency")
```

```
        smoker
sex       No Yes
  Female 258 241
  Male   270 231
```



1.3.3 Chi-square test:

This is a statistical test that can be used to determine if there is a significant association between two categorical variables. The test compares the observed frequencies in a contingency table to the expected frequencies under the null hypothesis of no association.

Suppose we have a dataset called survey that contains information on the smoking habits of students and faculty at a university. We want to test whether there is an association between smoking status (smoking_status) and gender (gender). We can do this using a chi-square test.

In [11]:

```
# Set seed for reproducibility
set.seed(123)

# Create a vector of smoking statuses
smoking_status <- sample(c("Non-smoker", "Occasional smoker", "Regular smoker

# Create a vector of genders
gender <- sample(c("Male", "Female"), size = 100, replace = TRUE)

# Create a data frame with the generated data
survey <- data.frame(smoking_status = smoking_status, gender = gender)

# Print the first few rows of the survey data frame
survey
```

A data.frame: 100 × 2

smoking_status	gender
<fct>	<fct>
Regular smoker	Male
Regular smoker	Female
Regular smoker	Female
Occasional smoker	Male
Regular smoker	Male
Occasional smoker	Male
Occasional smoker	Female
Occasional smoker	Male
Regular smoker	Female
Non-smoker	Female
Occasional smoker	Female
Occasional smoker	Female
Non-smoker	Female
Occasional smoker	Female
Regular smoker	Female
Non-smoker	Male
Regular smoker	Female
Regular smoker	Female
Non-smoker	Female
Non-smoker	Male
Non-smoker	Female
Non-smoker	Female
Regular smoker	Female
Occasional smoker	Male
Regular smoker	Female
Occasional smoker	Female
Non-smoker	Male
Occasional smoker	Female
Regular smoker	Male
Occasional smoker	Female
⋮	⋮
Occasional smoker	Female
Regular smoker	Male
Regular smoker	Male

smoking_status	gender
<fct>	<fct>
Non-smoker	Female
Occasional smoker	Male
Occasional smoker	Female
Non-smoker	Female
Occasional smoker	Female
Non-smoker	Female
Non-smoker	Male
Occasional smoker	Female
Regular smoker	Female
Regular smoker	Female
Non-smoker	Male
Occasional smoker	Female
Non-smoker	Female
Occasional smoker	Female
Non-smoker	Male
Regular smoker	Male
Regular smoker	Female
Occasional smoker	Female
Regular smoker	Female
Non-smoker	Female
Occasional smoker	Male
Occasional smoker	Female
Regular smoker	Male
Occasional smoker	Female
Non-smoker	Male
Regular smoker	Male
Regular smoker	Male

In [12]:
```
# Load required library
library(stats)

# Create contingency table
ct <- table(survey$smoking_status, survey$gender)

# Conduct chi-square test
chisq.test(ct)
```

```
        Pearson's Chi-squared test

data:  ct
X-squared = 1.7694, df = 2, p-value = 0.4128
```

If the p-value is less than the chosen significance level (e.g., 0.05), we can reject the null hypothesis and conclude that there is a significant association between smoking_status and gender. If the p-value is greater than the chosen significance level, we fail to reject the null hypothesis and conclude that there is not enough evidence to suggest an association between the two variables.

In the previous example, the significance level was not explicitly defined in the code. By default, the chisq.test() function in R uses a significance level of 0.05 for hypothesis testing.

1.3.4 Correspondence analysis:

This is a multivariate technique used to explore the relationships between categorical variables. It produces a graphical representation of the relationships between categories in a contingency table, which can help identify patterns and trends in the data.

In [3]:

```
# Set seed for reproducibility
set.seed(123)

# Create a vector of smoking statuses
smoking_status <- sample(c("Non-smoker", "Occasional smoker", "Regular smoker

# Create a vector of age groups
age_group <- sample(c("18-24", "25-34", "35-44", "45-54", "55+"), size = 100,

# Create a data frame with the generated data
survey <- data.frame(smoking_status = smoking_status, age_group = age_group)

# Print the first few rows of the survey data frame
head(survey)
```

A data.frame: 6 × 2

	smoking_status	age_group
	<fct>	<fct>
1	Regular smoker	25-34
2	Regular smoker	55+
3	Regular smoker	55+
4	Occasional smoker	35-44
5	Regular smoker	25-34
6	Occasional smoker	25-34

In [4]:

```
# Load required libraries
library(FactoMineR)
library(factoextra)

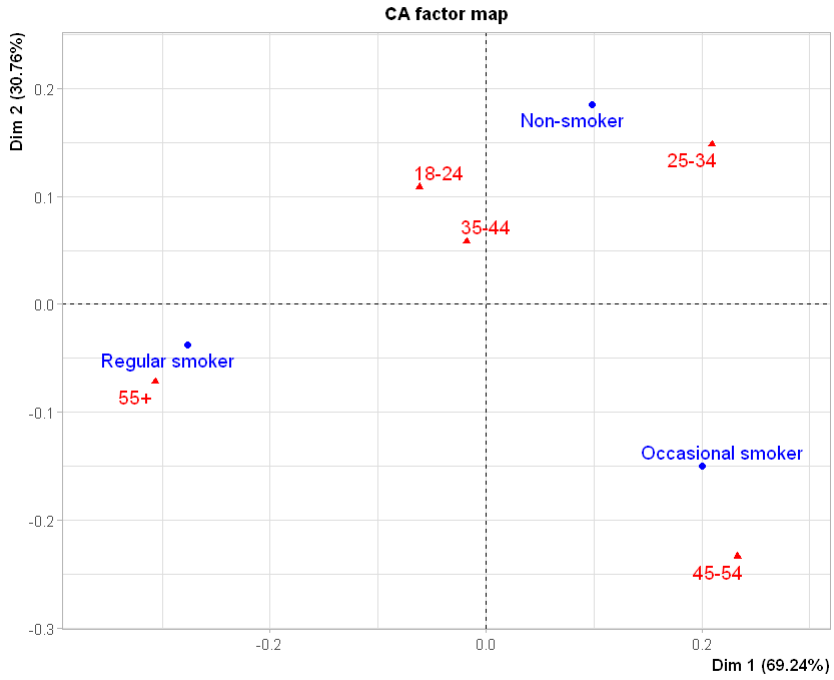
# Create contingency table
ct <- table(survey$smoking_status, survey$age_group)

ct

# Conduct correspondence analysis
ca <- CA(ct)

# Plot results
#fviz_ca(ca)
```

```
                  18-24 25-34 35-44 45-54 55+
Non-smoker            7     9     6     5   6
Occasional smoker     5     7     5     9   6
Regular smoker        7     5     6     5  12
```



1.4 Descriptive statistics in numerical data

1.4.1 Measures of central tendency

Mean:

- Add up all of the values to find a total.
- Divide the total by the number of values you added together.

In [2]:
```
v1 = c(2,9,1,5,3,7,8)
```

In [1]:
```
sum1 = sum(v1)
sum1

mean = sum1/length(v1)
mean
```

```
Error in eval(expr, envir, enclos): object 'v1' not found
Traceback:
```

At first glance, summarizing data might seem fairly trivial: just take the mean of the data (see “Mean”). In fact, while the mean is easy to compute and expedient to use, it may not always be the best measure for a central value. For this reason, statisticians have developed and promoted several alternative estimates to the mean.

A variation of the mean is a trimmed mean, which you calculate by dropping a fixed number of sorted values at each end and then taking an average of the remaining values. Representing the sorted values by x(1), x(2),...,x(n), where x(1) is the smallest value and x(n) the largest, the formula to compute the trimmed mean with P smallest and largest values omitted is:

$$\text{Trimmed mean} = \bar{x} = \frac{\sum_{i=p+1}^{n-p} x_{(i)}}{n-2p}$$

A trimmed mean eliminates the influence of extreme values. For example, in international diving the top and bottom scores from five judges are dropped, and the final score is the average of the three remaining judges

This makes it difficult for a single judge to manipulate the score, perhaps to favor his country’s contestant. Trimmed means are widely used, and in many cases, are preferable to use instead of the ordinary mean

Another type of mean is a weighted mean, which you calculate by multiplying each data value x_i bu a weight w_i and dividing their sum by the sum of the weights. The formula for a weighted mean is:

$$\text{Weighted mean} = \bar{x}_w = \frac{\sum_{i=1}^{n} w_i x_i}{\sum_{i}^{n} w_i}$$

There are two main motivations for using a weighted mean:

- Some values are intrinsically more variable than others, and highly variable observations are given a lower weight. For example, if we are taking the average from multiple sensors and one of the sensors is less accurate, then we might down weight the data from that sensor.
- The data collected does not equally represent the different groups that we are interested in measuring. For example, because of the way an online experiment was conducted, we may not have a set of data that accurately reflects all groups in the user base. To correct that, we can give a higher weight to the values from the groups that were underrepresented.

Median

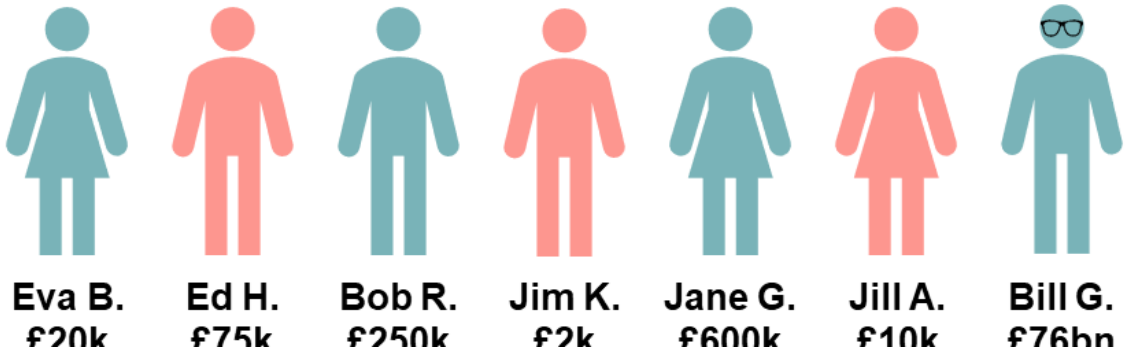
Is the middle value of ordered data

- Put all of the values into order.
- The median if the middle value.
- If there are two values in the middle, find the mean of these two values.

In [ ]:
```
median(v1)
```

The median is the middle number on a sorted list of the data. If there is an even number of data values, the middle value is one that is not actually in the data set, but rather the average of the two values that divide the sorted data into upper and lower halves. Compared to the mean, which uses all observations, the median depends only on the values in the center of the sorted data. While this might seem to be a disadvantage, since the mean is much more sensitive to the data, there are many instances in which the median is a better metric for location.

The median is referred to as a robust estimate of location since it is not influenced by outliers (extreme cases) that could skew the results. An outlier is any value that is very distant from the other values in a data set. The exact definition of an outlier is somewhat subjective, although certain conventions are used in various data summaries and plots (“Percentiles and Boxplots”). Being an outlier in itself does not make a data value invalid or erroneous.



In [3]:
```
incomes <- c(20000, 75000, 250000, 2000, 600000, 10000, 76000000000000)
incomes
incomes_sorted <- sort(incomes)

incomes_sorted

print(paste("The mean is: ", mean(incomes_sorted)))

print(paste("The median is: ", median(incomes_sorted)))
```

```
20000  75000  250000  2000  6e+05  10000  7.6e+13

2000  10000  20000  75000  250000  6e+05  7.6e+13

[1] "The mean is:  10857142993857.1"
[1] "The median is:  75000"
```

Still, outliers are often the result of data errors such as mixing data of different units (kilometers versus meters) or bad readings from a sensor. When outliers are the result of bad data, the mean will result in a poor estimate of location, while the median will be still be valid. In any case, outliers should be identified and are usually worthy of further investigation.

Mode

Is the most frequen value.

- Count how many of each value appears.
- The mode is the value that appears the most.
- You can have more than one mode.

R does not have a standard in-built function to calculate mode. So we create a user function to calculate mode of a data set in R. This function takes the vector as input and gives the mode value as output.

Unike mean and median, mode can have both numeric and character data.

In [4]:
```
# Create the function.
getmode <- function(v) {
   uniqv <- unique(v)
   uniqv[which.max(tabulate(match(v, uniqv)))]
}

# Create the vector with numbers.
v <- c(2,1,2,3,1,2,3,4,1,5,5,3,2,3)

# Calculate the mode using the user function.
result <- getmode(v)
print(result)

# Create the vector with characters.
charv <- c("o","it","the","it","it")

# Calculate the mode using the user function.
result <- getmode(charv)
print(result)
```

```
[1] 2
[1] "it"
```

Range

Is the difference between the lowest and highest value

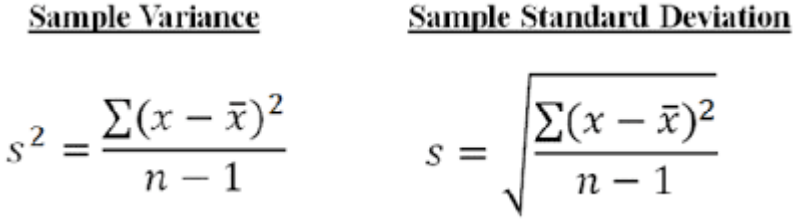
- Find the highest and lowest values.
- Substract the lowest values from the highest.

In [ ]:
```
range(v1)
```

1.4.2 Measures of dispersion

Standard deviation:

- Calculate the **mean** of the sample.
- Find the difference each entry (x) and the mean.
- Square the deviations from the mean.
- Sum the squares of the deviations from the mean.
- Divide the sum by (n-1) to get the **variance**.
- Take the square root of the variance to get the **standard deviation**.

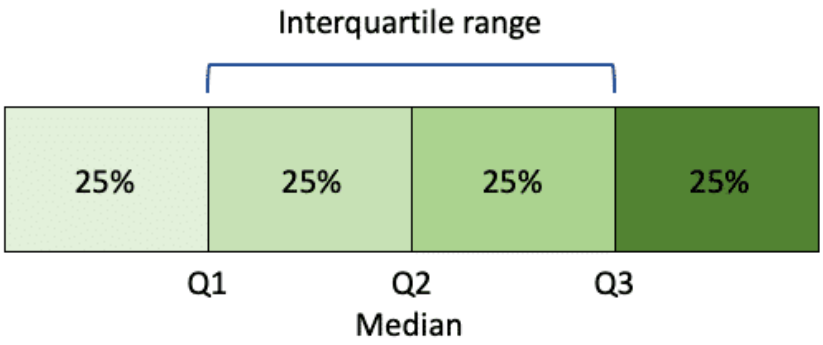


In [ ]:
```
sd(v1)
```

Interquartil Range (IQR):

The interquartile range tells you the spread of the middle half of your distribution.

Quartiles segment any distribution that's ordered from low to high into four equal parts. The interquartile range (IQR) contains the second and third quartiles, or the middle half of your data set.



IQR = Q3 - Q1

2 Bivariate analyses

The dividing line between a univariate and a bivariate study lies in the fact that the latter strictly seeks to analyze two variables together, which we could well denote as "X" and "Y" seeking to prove simple relationships of causality or association.

In statistics, a specific variant of bivariate studies is called **"correlational analysis"**, through which it is sought to demonstrate that there is a significant direct or inverse relationship between two variables.

There is a direct correlation between two variables when an increase in the "control variable" (independent) causes a similar change in the "outcome variable" (dependent).

In bivariate studies, discrimination between dependent and independent variables is not a mandatory requirement.

With a correlation analysis we test hypotheses that we might think as obvious as that a student's grade point average (dependent variable or outcome) is correlated with the time spent studying (independent or predictor).

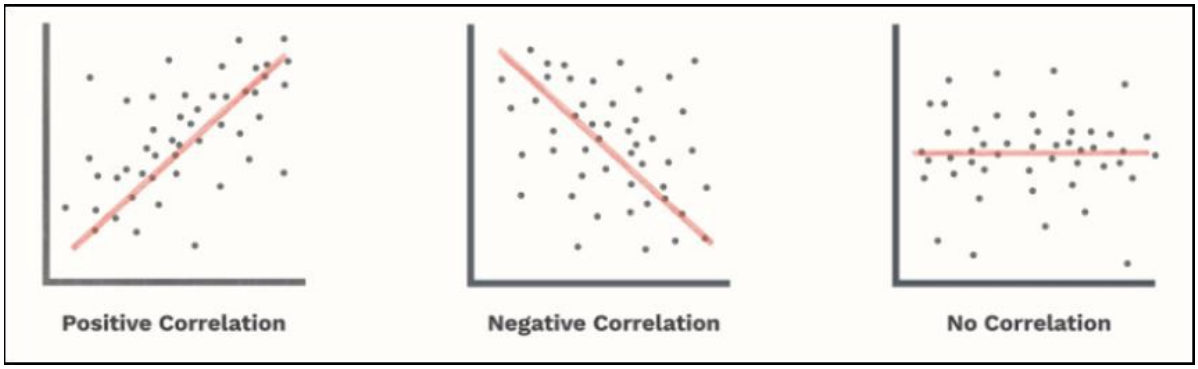
2.1 Coefficient of determination

A coefficient or statistical ratio that measures the degree of correlation is the coefficient of determination or “R squared”.

Coefficient of Correlation (r):

It measures the strength and the direction of a linear relationship between two variables (x and y) with possible values between -1 and 1.

- Positive Correlation: It indicates that two variables are in perfect harmony. They rise and fall together. +1 is perfect
- Negative Correlation: It indicates that two variables are perfect opposites. One goes up and other goes down. -1 is perfect
- No correlation: If there is no linear correlation or a weak linear correlation, r is close to 0.

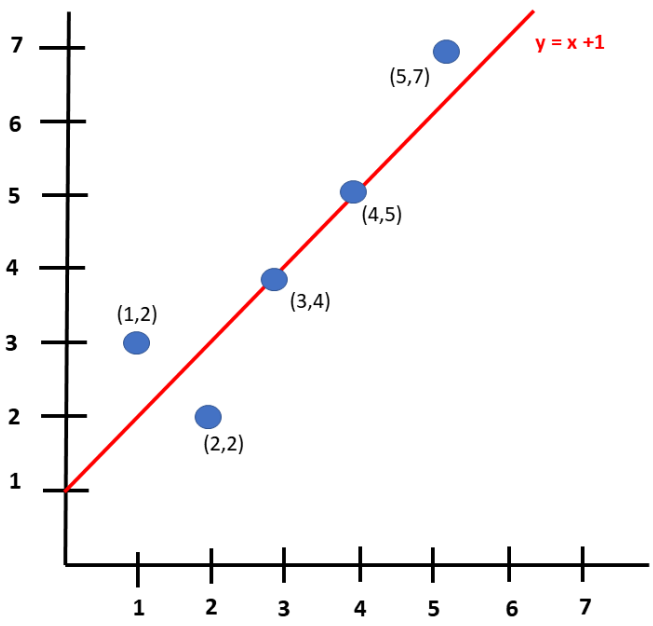


Coefficient of determination (R^2 or r^2):

$$r^2 = \frac{\text{RSS}}{\text{TSS}} = \frac{\sum(\hat{Y} - \bar{Y})^2}{\sum(Y - \bar{Y})^2}$$

We will use a equation to describe the academic performance of the students, where X are the hours of study and Y is the is the academic qualification obtained

Y = X + 1

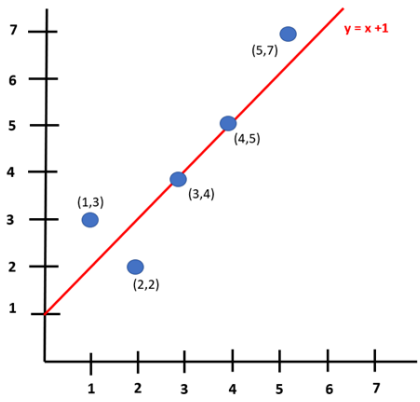


The closer the line is to the points, the better the prediction.

Two aspects to consider:

1. how well is the line (model) predicting the values
2. how different is the predicted value vs the actual value = Residual

$$r^2 = \frac{\text{RSS}}{\text{TSS}} = \frac{\sum(\hat{Y} - \bar{Y})^2}{\sum(Y - \bar{Y})^2}$$



Actual data (Y)

Y	X
3	1
2	2
4	3
5	4
7	5

Predicted data (Ŷ)

Ŷ	X
2	1
3	2
4	3
5	4
6	5

$$\bar{Y} = \frac{(3+2+4+5+7)}{5} = 4.2$$

We need to tabulate the actual and predicted data vs X. and then calculate the mean of actual data.

$$r^2 = \frac{\text{RSS}}{\text{TSS}} = \frac{\sum(\hat{Y} - \bar{Y})^2}{\sum(Y - \bar{Y})^2}$$

$\hat{Y}$	$\hat{Y} - 4.2$	$(\hat{Y} - 4.2)^2$
2	-2.2	4.84
3	-1.2	1.44
4	-0.2	0.04
5	0.8	0.64
6	1.8	3.24

$$\sum(\hat{Y} - \bar{Y})^2 = 10.2$$

Y	Y – 4.2	(Y – 4.2)²
3	-1.2	1.44
2	-2.2	4.84
4	-0.2	0.04
5	0.8	0.64
7	2.8	7.84

$$\sum(Y - \bar{Y})^2 = 14.8$$

$$\frac{\sum(\hat{Y} - \bar{Y})^2 = 10.2}{\sum(Y - \bar{Y})^2 = 14.8}$$

$r^2 = 0.6892$

As conclusion:

The 68.92% of the variance of school scores can be explained due to the hours of study

Finding the coefficient of determination for the simple linear regression model of the data set faithful.

In [5]:
```
head(faithful)
```

eruptions	waiting
3.600	79
1.800	54
3.333	74
2.283	62
4.533	85
2.883	55

In [6]:
```
eruption.lm = lm(eruptions ~ waiting, data=faithful)
```

In [7]:
```
summary(eruption.lm)
```

```
Call:
lm(formula = eruptions ~ waiting, data = faithful)

Residuals:
     Min       1Q   Median       3Q      Max
-1.29917 -0.37689  0.03508  0.34909  1.19329

Coefficients:
             Estimate Std. Error t value Pr(>|t|)
(Intercept) -1.874016   0.160143  -11.70   <2e-16 ***
waiting      0.075628   0.002219   34.09   <2e-16 ***
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 0.4965 on 270 degrees of freedom
Multiple R-squared:  0.8115,    Adjusted R-squared:  0.8108
F-statistic:  1162 on 1 and 270 DF,  p-value: < 2.2e-16
```

In [8]:
```
summary(eruption.lm)$r.squared


print(paste0("R^2= ", summary(eruption.lm)$r.squared))
```

0.81146076097331

```
[1] "R^2= 0.81146076097331"
```

2.2 Covariance

In statistics, the covariance is a measure of the join variability of two random variables.

$$\mathbf{VAR(X)} = \frac{1}{n-1}\sum_{i=1}^{n}(x_i - E(X))^2 \tag{1}$$

$$\mathbf{COV(X,\ Y)} = \frac{1}{n-1}\sum_{i=1}^{n}(x_i - E(X))(y_i - E(Y)) \tag{2}$$

$$\mathbf{COR(X,\ Y)} = \frac{\mathbf{COV(X,\ Y)}}{\sqrt{\mathbf{VAR(X)VAR(Y)}}} \tag{3}$$

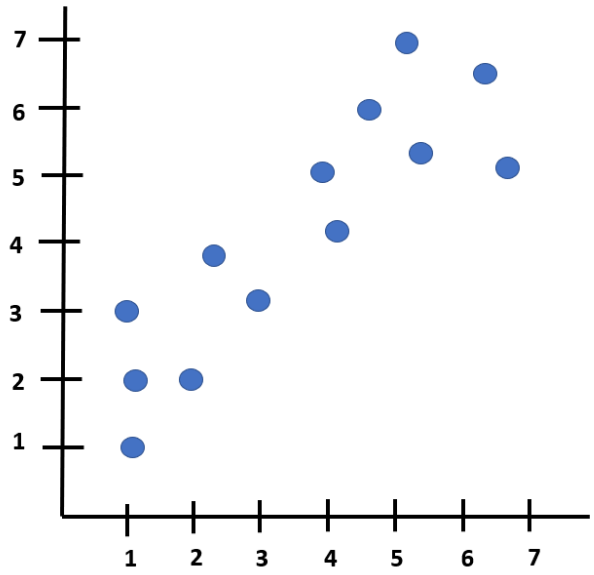
$$\mathbf{R}^2 = 1 - \frac{\mathbf{VAR(X,\ Y)}_{FittedLine}}{\mathbf{VAR(X,\ Y)}_{Mean}} \tag{4}$$

Covariance

$$\sigma_{xy} = \frac{\sum(x_i - \overline{x})(y_i - \overline{y})\, n_i}{N}$$

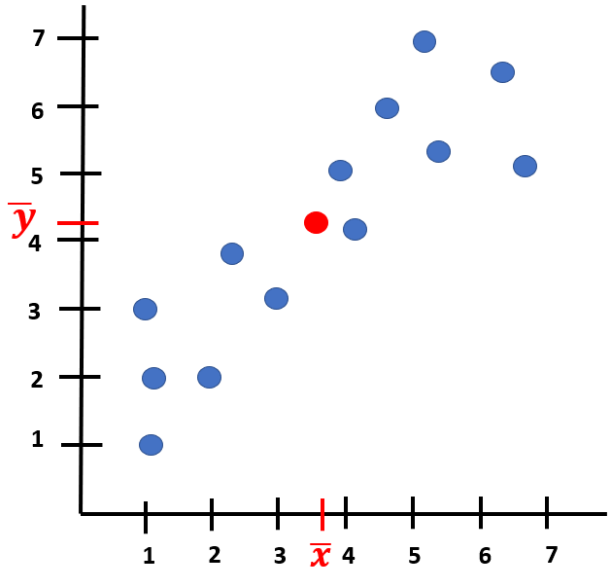
$$\sigma_{xy} = \frac{\sum x_i y_i \; n_i}{N} - \overline{x}\,\overline{y}$$

Now, we will analyse positive covariance between school scores and hours of study.



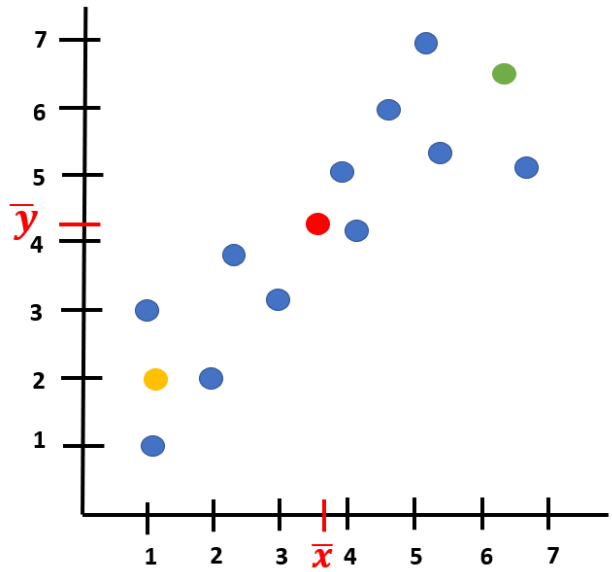
$$\sigma_{xy} = \frac{\sum(x_i - \overline{x})(y_i - \overline{y})\, n_i}{N}$$

lets plot the average in x and y



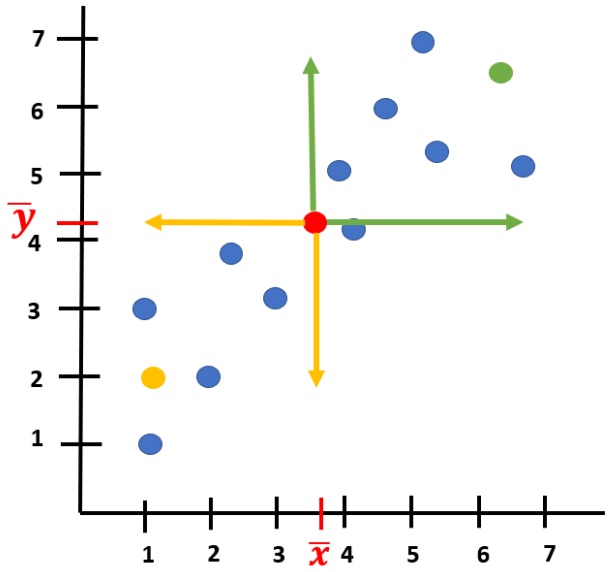
$$\sigma_{xy} = \frac{\sum(x_i - \overline{x})(y_i - \overline{y})\, n_i}{N}$$

If we select one random value (x,y)



$$\sigma_{xy} = \frac{\sum(x_i - \overline{x})(y_i - \overline{y})\, n_i}{N}$$

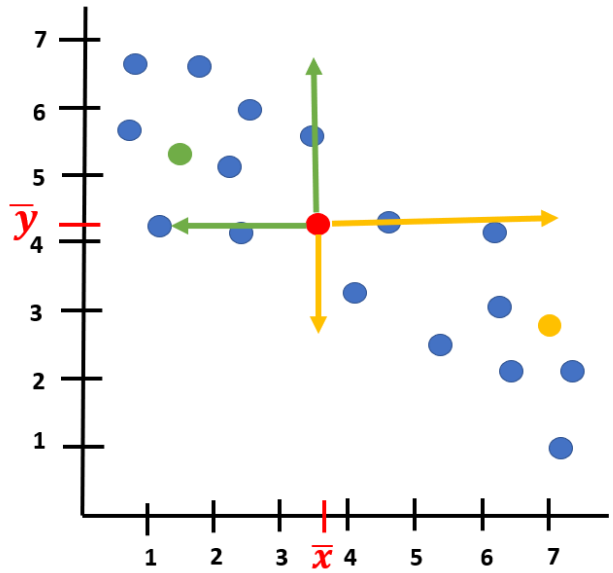
How are the yellow and green points positioned with respect to the mean of x and y?



$$\sigma_{xy} = \frac{\sum(x_i - \overline{x})(y_i - \overline{y})\, n_i}{N} > 0$$

$$\sigma_{xy} = \frac{\sum(x_i - \overline{x})(y_i - \overline{y})\, n_i}{N} > 0$$

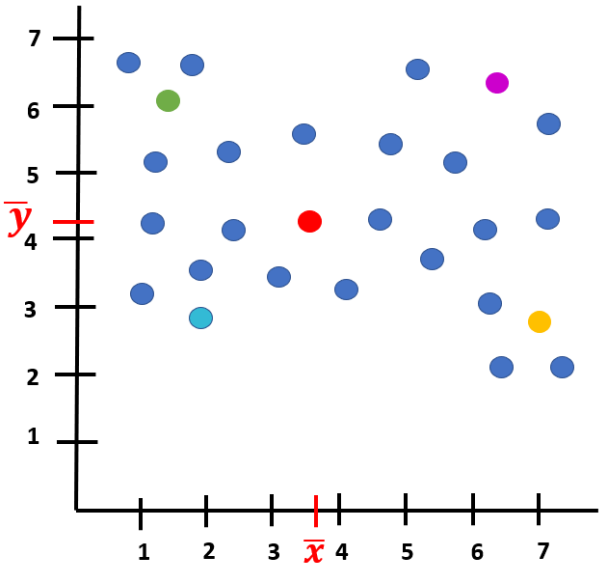
When there is a direct relationship between two variables, the covariance gives a positive value.



$$\sigma_{xy} = \frac{\sum(x_i - \overline{x})(y_i - \overline{y})\, n_i}{N} < 0$$

$$\sigma_{xy} = \frac{\sum(x_i - \overline{x})(y_i - \overline{y})\, n_i}{N} < 0$$

When there is a inverse relationship between two variables, the covariance gives a negative value.



$$\sigma_{xy} = \frac{\sum(x_i - \overline{x})(y_i - \overline{y})\, n_i}{N}$$

$$\sigma_{xy} = \frac{\sum(x_i - \overline{x})(y_i - \overline{y})\, n_i}{N}$$

$$\sigma_{xy} = \frac{\sum(x_i - \overline{x})(y_i - \overline{y})\, n_i}{N}$$

$$\sigma_{xy} = \frac{\sum(x_i - \overline{x})(y_i - \overline{y})\, n_i}{N}$$

When there is NOT a relationship between two variables, the covariance gives a value near to zero.

In [9]:
```
library(gapminder)
```

```
Warning message:
"package 'gapminder' was built under R version 3.6.3"
```

In [10]:
```
summary(gapminder)
```

```
       country        continent        year         lifeExp     
 Afghanistan:  12   Africa  :624   Min.   :1952   Min.   :23.60  
 Albania    :  12   Americas:300   1st Qu.:1966   1st Qu.:48.20  
 Algeria    :  12   Asia    :396   Median :1980   Median :60.71  
 Angola     :  12   Europe  :360   Mean   :1980   Mean   :59.47  
 Argentina  :  12   Oceania : 24   3rd Qu.:1993   3rd Qu.:70.85  
 Australia  :  12                  Max.   :2007   Max.   :82.60  
 (Other)    :1632                                                
      pop              gdpPercap       
 Min.   :6.001e+04   Min.   :   241.2  
 1st Qu.:2.794e+06   1st Qu.:  1202.1  
 Median :7.024e+06   Median :  3531.8  
 Mean   :2.960e+07   Mean   :  7215.3  
 3rd Qu.:1.959e+07   3rd Qu.:  9325.5  
 Max.   :1.319e+09   Max.   :113523.1  
```

In [11]:
```
cov(gapminder$pop, gapminder$year)
```

```
150858448.310335
```

In [12]:
```
duration = faithful$eruptions   # eruption durations
waiting = faithful$waiting      # the waiting period
cov(duration, waiting)
```

13.9778078467549

2.3 Pearson's correlation coefficient

What problem can we see when comparing the covariance of both data shown above? 13.977 vs 150858448.310

Pearson coefficient

$$r = \frac{\sigma_{xy}}{\sigma_x \ \sigma_y}$$

The Pearson parameter allow us to compare between different measures since it has no units! because is a coefficient

-1 ≤ r ≤ 1

If r ≈ 1, then there exist a strong direct correlation

If r ≈ -1, then there exist a strong inverse correlation

If r = -1 or r = 1, then there exist a function correlation where it is possible to find a mathematical function to describe the relationship

If r ≈ 0, then there is not exist a lineal correlation

In [13]:

```
cor.test(faithful$eruptions, faithful$waiting,
                     method = "pearson")


cor.test(gapminder$pop, gapminder$year,
                     method = "pearson")
```

```
        Pearson's product-moment correlation

data:  faithful$eruptions and faithful$waiting
t = 34.089, df = 270, p-value < 2.2e-16
alternative hypothesis: true correlation is not equal to 0
95 percent confidence interval:
 0.8756964 0.9210652
sample estimates:
      cor
0.9008112


        Pearson's product-moment correlation

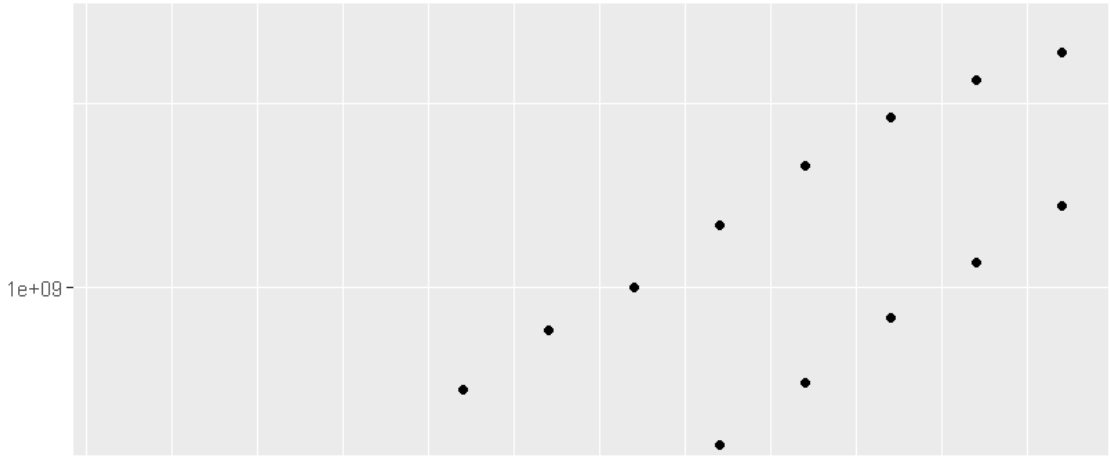
data:  gapminder$pop and gapminder$year
t = 3.4072, df = 1702, p-value = 0.0006716
alternative hypothesis: true correlation is not equal to 0
95 percent confidence interval:
 0.03495832 0.12928914
sample estimates:
       cor
0.08230808
```

In [14]:
```
library(ggplot2)

ggplot(gapminder, aes(x=year, y=pop))+
geom_point()

ggplot(faithful, aes(x=eruptions, y=waiting))+
geom_point()
```

```
Registered S3 methods overwritten by 'ggplot2':
  method         from
  [.quosures     rlang
  c.quosures     rlang
  print.quosures rlang
```



In [15]:
```
library(dplyr)

gapminder_year <- gapminder %>%
group_by(year) %>%
summarise(totalpop = sum(as.numeric(pop)))

gapminder_year
```

```
Attaching package: 'dplyr'

The following objects are masked from 'package:stats':

    filter, lag

The following objects are masked from 'package:base':

    intersect, setdiff, setequal, union
```

year	totalpop
1952	2406957150
1957	2664404580
1962	2899782974
1967	3217478384
1972	3576977158
1977	3930045807
1982	4289436840
1987	4691477418
1992	5110710260
1997	5515204472
2002	5886977579
2007	6251013179

In [16]:
```
#Covariance faithful
cov(faithful$eruptions, faithful$waiting)

#Covariance gapminder modified
cov(gapminder_year$totalpop, gapminder_year$year)

#Pearson faithful
cor.test(faithful$eruptions, faithful$waiting,
                    method = "pearson")

#Pearson gapminder modified
cor.test(gapminder_year$totalpop, gapminder_year$year,
                    method = "pearson")
```

13.9778078467549

23355630679.3182

```
        Pearson's product-moment correlation

data:  faithful$eruptions and faithful$waiting
t = 34.089, df = 270, p-value < 2.2e-16
alternative hypothesis: true correlation is not equal to 0
95 percent confidence interval:
 0.8756964 0.9210652
sample estimates:
      cor
0.9008112
```

```
        Pearson's product-moment correlation

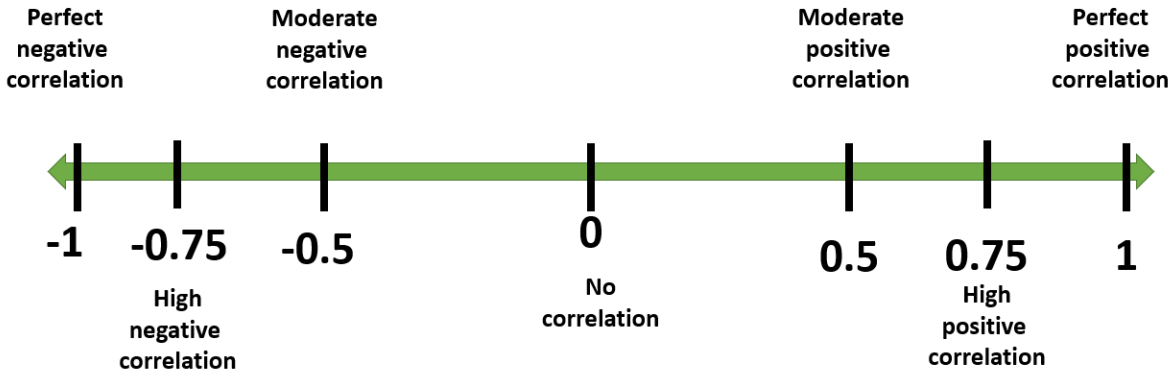
data:  gapminder_year$totalpop and gapminder_year$year
t = 45.666, df = 10, p-value = 6.104e-13
alternative hypothesis: true correlation is not equal to 0
95 percent confidence interval:
 0.9912038 0.9993527
sample estimates:
     cor
0.997611
```

Now both Pearson coefficients are comparable

2.4 Spearman's rho coefficient

Rho of Spearman

$$r_s = 1 - \frac{6\sum D^2}{n\left(n^2 - 1\right)}$$



	A	B	C	D	E	F	G	H	I	J
1	Student	Mathematics (X)	Statistics (Y)	rango (x)	rango (y)	d	d2			
2	18	6	7	1,0	2,5	1,50	2,25		n=	20
3	2	7	13	2,5	11,0	8,50	72,25			
4	17	7	6	2,5	1,0	-1,50	2,25		Σd2=	403
5	1	8	12	4,5	9,5	5,00	25			
6	10	8	9	4,5	5,5	1,00	1		ρ =	0,69699
7	9	9	7	6,0	2,5	-3,50	12,25			
8	8	10	12	7,0	9,5	2,50	6,25			
9	11	11	19	8,5	16,5	8,00	64			
10	12	11	8	8,5	4,0	-4,50	20,25			
11	3	12	10	10,0	7,5	-2,50	6,25			
12	6	13	14	11,0	12,5	1,50	2,25			
13	7	14	14	12,5	12,5	0,00	0			
14	20	14	9	12,5	5,5	-7,00	49			
15	16	15	19	14,0	16,5	2,50	6,25			
16	13	16	19	15,0	16,5	1,50	2,25			
17	4	17	19	16,0	16,5	0,50	0,25			
18	5	18	18	17,0	14,0	-3,00	9			
19	14	19	20	18,5	19,5	1,00	1			
20	19	19	10	18,5	7,5	-11,00	121			
21	15	20	20	20,0	19,5	-0,50	0,25			
22						Sum	403			
23										

In [18]:
```
#Pearson gapminder modified
cor.test(gapminder_year$totalpop, gapminder_year$year,
                    method = "pearson")

#Spearman gapminder modified
cor.test(gapminder_year$totalpop, gapminder_year$year,
                    method = "spearman")
```

```
        Pearson's product-moment correlation

data:  gapminder_year$totalpop and gapminder_year$year
t = 45.666, df = 10, p-value = 6.104e-13
alternative hypothesis: true correlation is not equal to 0
95 percent confidence interval:
 0.9912038 0.9993527
sample estimates:
     cor
0.997611


        Spearman's rank correlation rho

data:  gapminder_year$totalpop and gapminder_year$year
S = 0, p-value < 2.2e-16
alternative hypothesis: true rho is not equal to 0
sample estimates:
rho
  1
```

You should use Spearman's Rho in the following scenario:

- You want to know the relationship between two variables
- Your variables of interest are continuous with outliers or **ordinal**
- You have only two variables

Variable 1	Variable 2	Method to test for correlation (dependence)
Numerical	Numerical	Pearson correlation
Categorical (Ordinal)	Numerical	Spearman's rank correlation

In [19]:
```
directions <- c("North", "East", "South", "West")
x<-factor(directions, levels= c("North", "East", "South", "West"))

cities <- c("New York","Rome","Paris","London")
y<-factor(cities, levels= c("New York","Rome","Paris","London"))

directions
x

cities
y


cor(as.numeric(x), as.numeric(y), method="spearman")
cor.test(as.numeric(x), as.numeric(y), method="spearman")
```

'North' 'East' 'South' 'West'

North East South West

▶ **Levels**:

'New York' 'Rome' 'Paris' 'London'

New York Rome Paris London

▶ **Levels**:

1

```
        Spearman's rank correlation rho

data:  as.numeric(x) and as.numeric(y)
S = 0, p-value = 0.08333
alternative hypothesis: true rho is not equal to 0
sample estimates:
rho
  1
```

3 Multivariate analysis

In a multivariate analysis, we do not speak of "simple correlation" or descriptive statistics by themselves, but rather we use other statistical tools called "multivariate", such as:

- Analysis of variance (ANOVA)
- Multiple variance analysis
- Principal component analysis (PCA)
- Trajectory analysis
- Multifactorial study
- Multiple regression analyses

3.1 p-value

What is a hypothesis in statistics?

It is a question that admits a **yes** or **no** answer.

For example:

Is the average salary in Chile higher for men than for women? ----> it is a **comparison**

Helicobacter pylori infection is associated with gastric cancer? ---> it is a **relation or association**

The number of hours of autonomous study is the cause of success in a doctoral program? ----> it is a **causality**

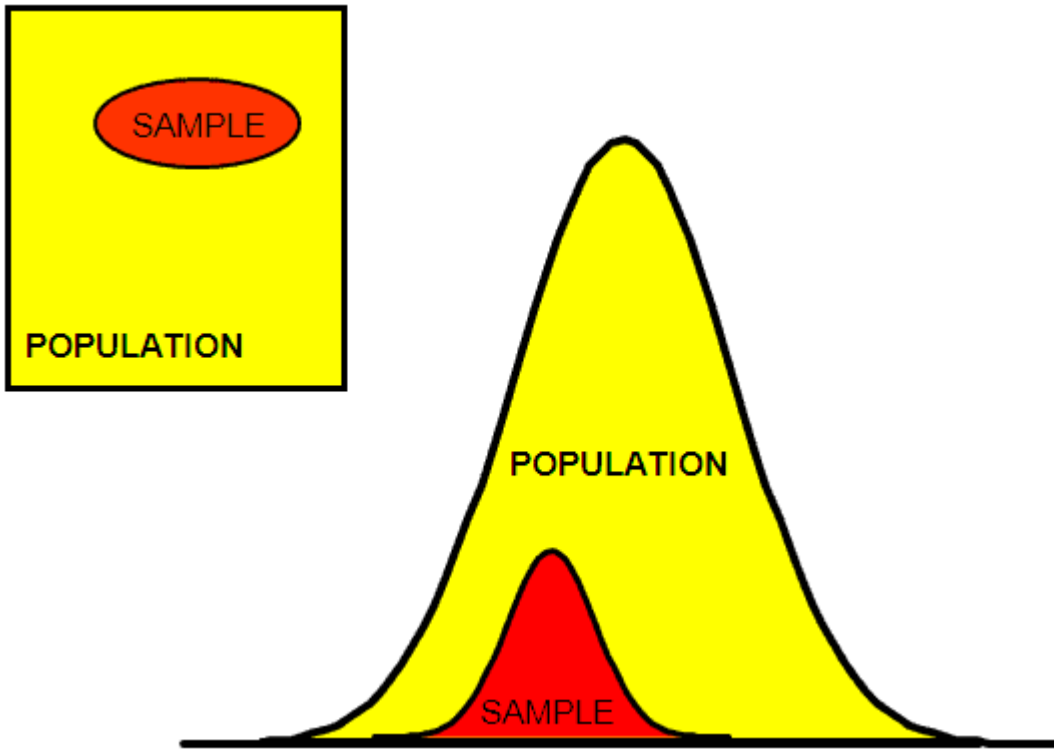
The questions posed before can be answered with the **p-value**

Is the average salary in Chile higher for men than for women?

So, we have two options:

H1: average salary is **NOT equal** between men and women.

H0: average salary is **equal** between men and women.



The p-value is the error probability of accepting H1 as true and actually being false

Is the average salary in Chile higher for men than for women?

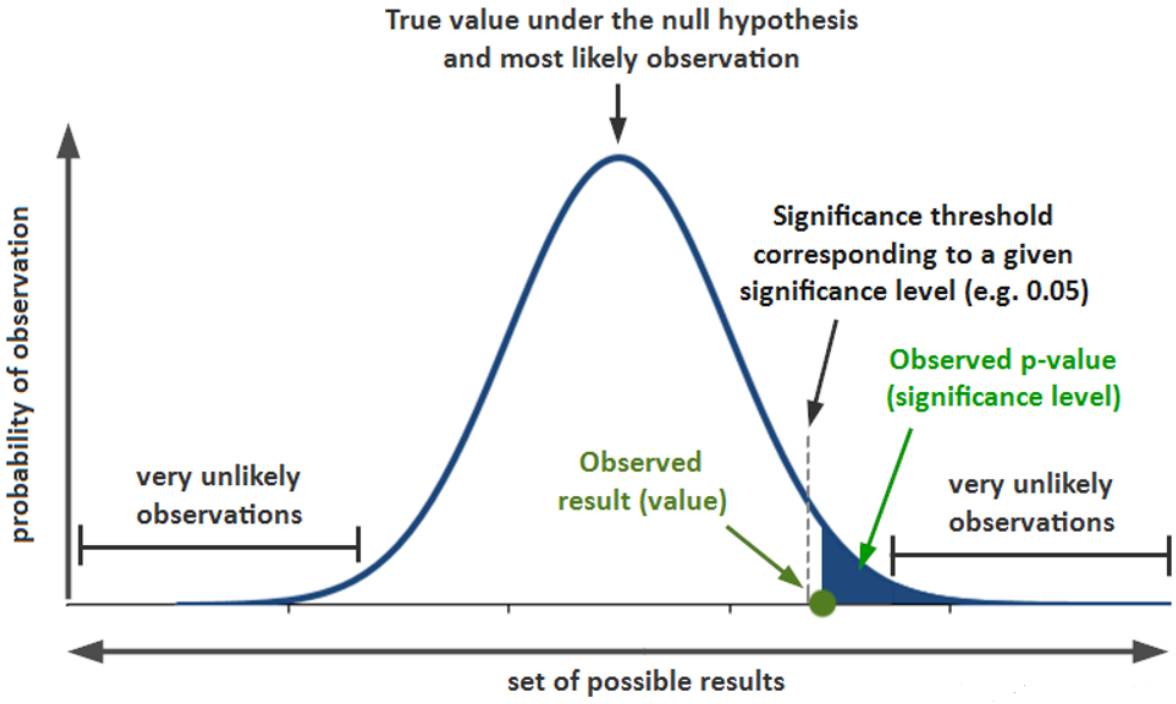
So, we have two options:

H1: average salary is **NOT equal** between men and women.

H0: average salary is **equal** between men and women.

p-value = 0.08 ----> 8 out of 100 times we will accept differences between salaries and in reality IT IS NOT TRUE

What about p-value = 0.08 vs p-value = 0.03? Which generates more confidence (less error)?



The p-value means that the probability of the null hypothesis will be true

When:

p < 0.05 : The null hypotesis (H0) is Falsa / The alternative hypothesis (H1) is True

p > 0.05 : The null hypothesis (H0) is True

The smaller the p-value, the more reliable the study result.

3.2 ANOVA

An "Analysis of Variance" (ANOVA) tests three or more groups for mean differences based on a continuous (i.e. scale or interval) response variable (a.k.a. dependent variable). The term "factor" refers to the variable that distinguishes this group membership. Race, level of education, and treatment condition are examples of factors.

There are two main types of ANOVA: (1) "one-way" ANOVA compares levels (i.e. groups) of a single factor based on single continuous response variable (e.g. comparing test score by 'level of education') and (2) a "two-way" ANOVA compares levels of two or more factors for mean differences on a single continuous response variable(e.g. comparing test score by both 'level of education' and 'zodiac sign').

The ANOVA test allows a comparison of more than two groups at the same time to determine whether a relationship exists between them. The result of the ANOVA formula, the F statistic (also called the F-ratio), allows for the analysis of multiple groups of data to determine the variability between samples and within samples.

If no real difference exists between the tested groups, which is called the null hypothesis, the result of the ANOVA's F-ratio statistic will be close to 1.

The Formula for ANOVA is:

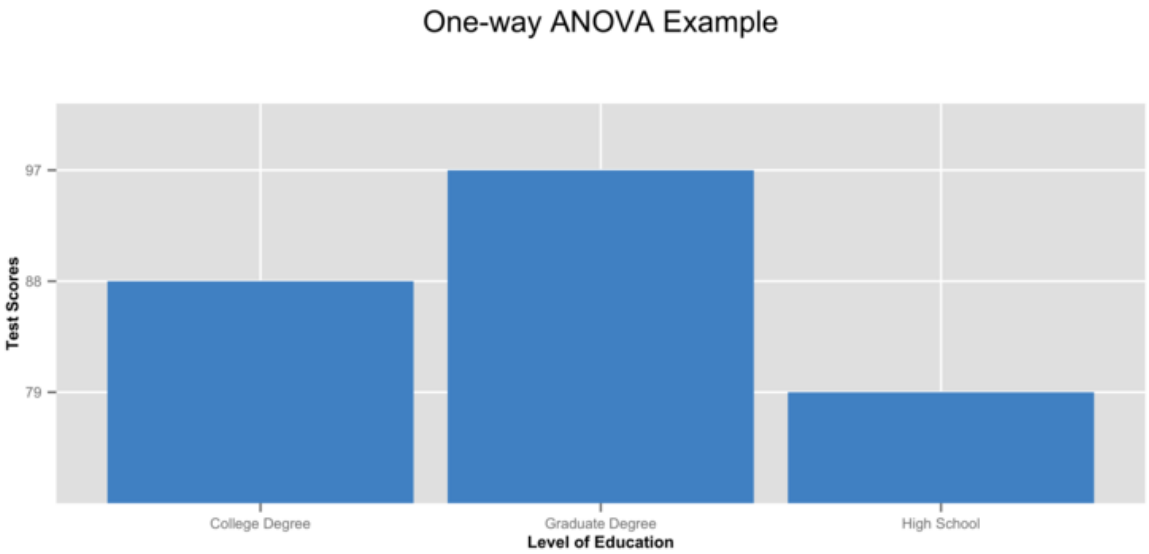
$$F = \frac{MST}{MSE}$$

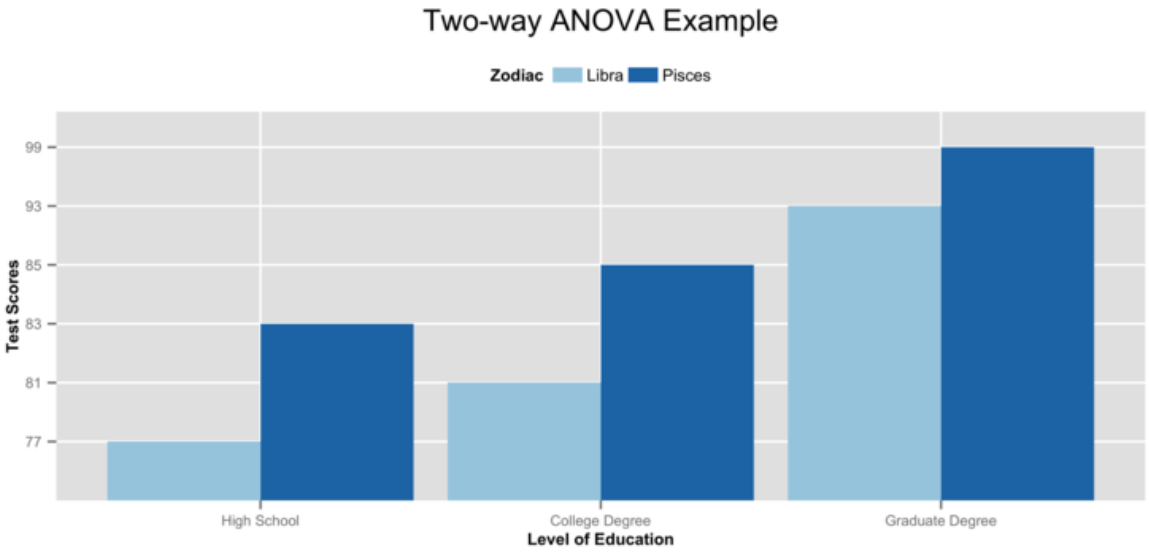
where:

$F = ANOVA\ coefficient$

$MST = Mean\ sum\ of\ squares\ due\ to\ treatment$

$MSE = Mean\ sum\ of\ squares\ due\ to\ error$

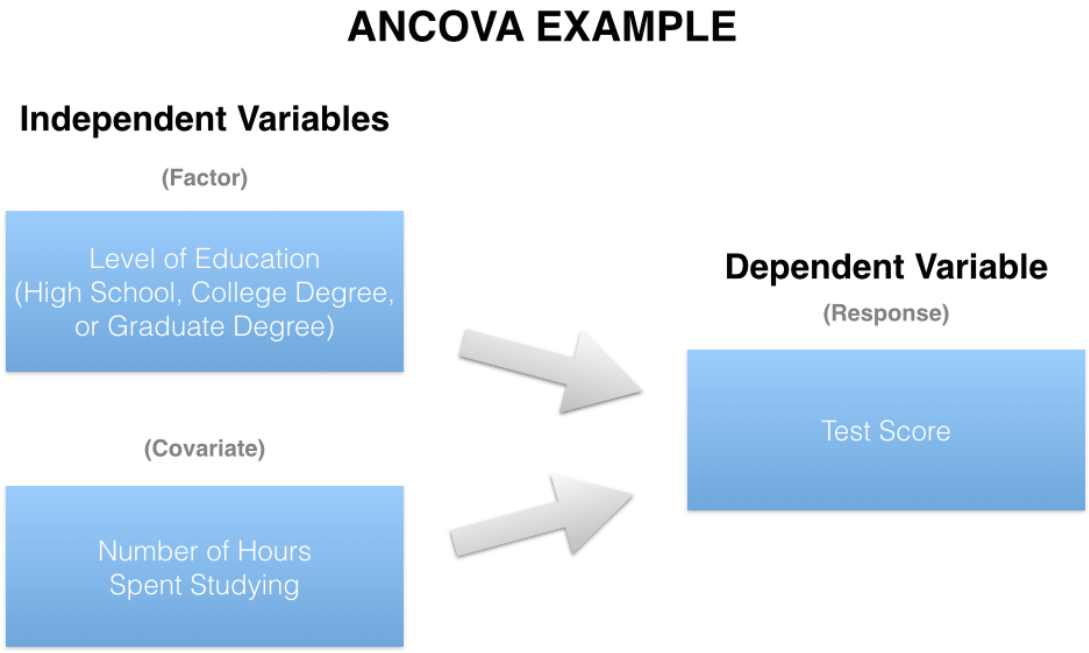




3.3 ANCOVA

The obvious difference between ANOVA and ANCOVA is the the letter "C", which stands for 'covariance'. Like ANOVA, "Analysis of Covariance" (ANCOVA) has a single continuous response variable. Unlike ANOVA, ANCOVA compares a response variable by both a factor and a continuous independent variable (e.g. comparing test score by both 'level of education' and 'number of hours spent studying'). The term for the continuous independent variable used in ANCOVA is "covariate".

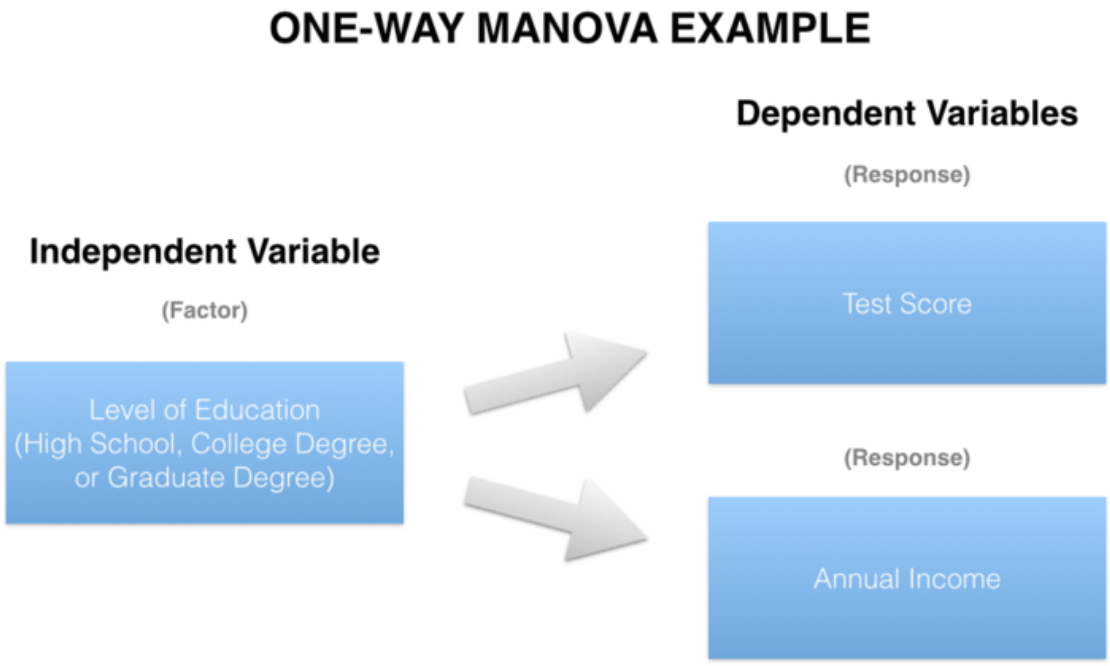
ANCOVA is also commonly used to describe analyses with a single response variable, continuous IVs, and no factors. Such an analysis is also known as a regression.

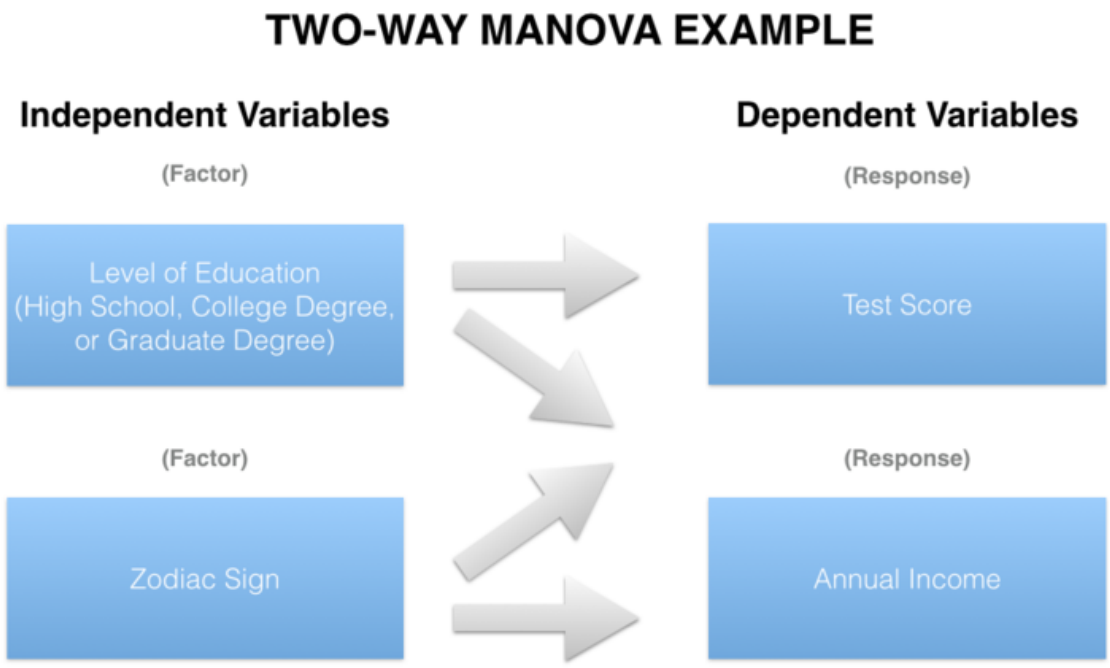


3.4 MANOVA

The obvious difference between ANOVA and a "Multivariate Analysis of Variance" (MANOVA) is the “M”, which stands for multivariate. In basic terms, A MANOVA is an ANOVA with two or more continuous response variables. Like ANOVA, MANOVA has both a one-way flavor and a two-way flavor. The number of factor variables involved distinguish a one-way MANOVA from a two-way MANOVA.

When comparing two or more continuous response variables by a single factor, a one-way MANOVA is appropriate (e.g. comparing ‘test score’ and ‘annual income’ together by ‘level of education’). A two-way MANOVA also entails two or more continuous response variables, but compares them by at least two factors (e.g. comparing ‘test score’ and ‘annual income’ together by both ‘level of education’ and ‘zodiac sign’).

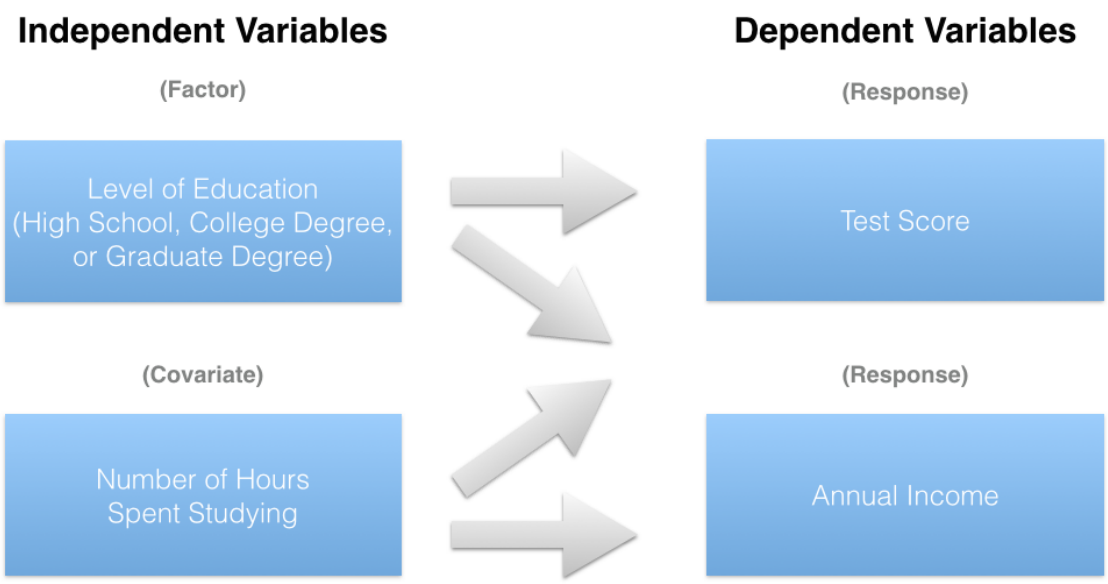




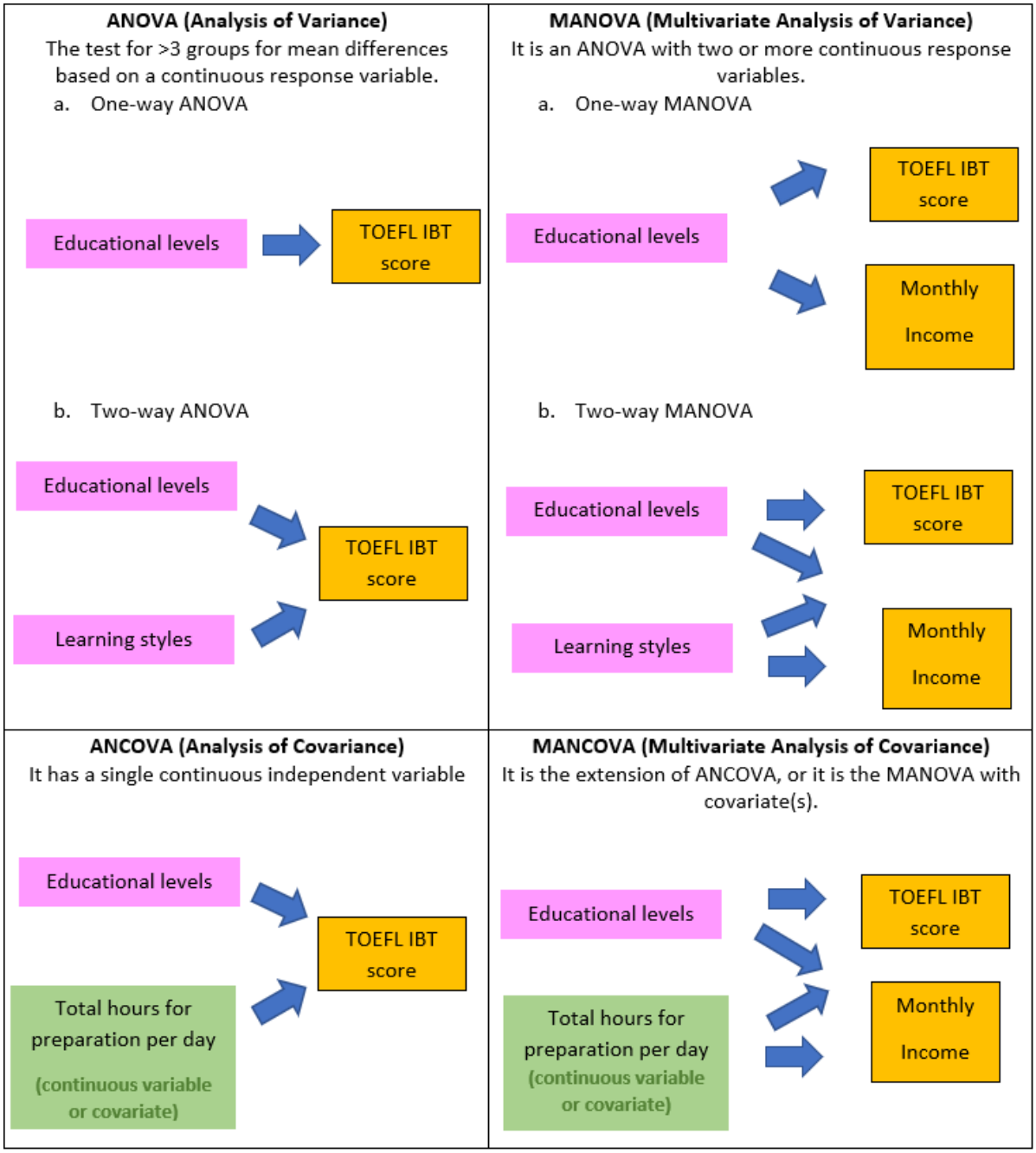
3.5 MANCOVA

Like ANOVA and ANCOVA, the main difference between MANOVA and MANCOVA is the “C,” which again stands for “covariance.” Both a MANOVA and MANCOVA feature two or more response variables, but the key difference between the two is the nature of the Independent variables. While a MANOVA can include only factors, an analysis evolves from MANOVA to MANCOVA when one or more covariates are added to the mix.

MANCOVA EXAMPLE



3.6 Examples ANOVA



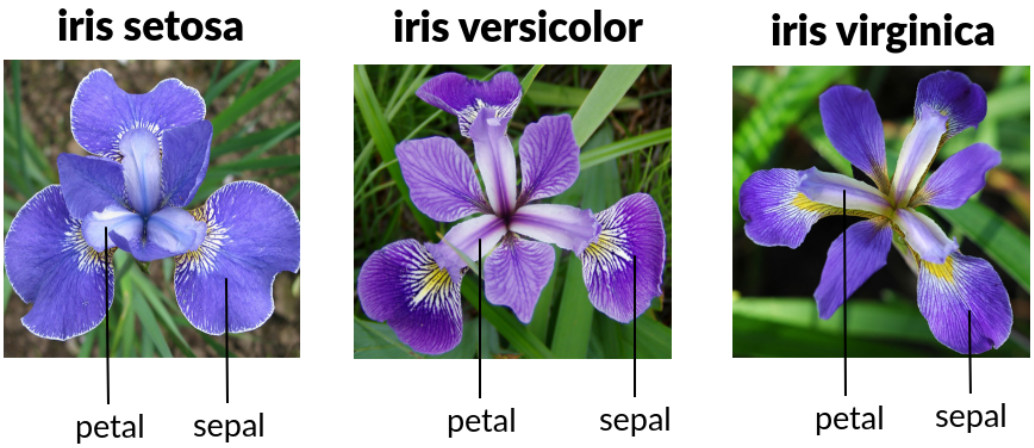
3.6.1 The IRIS data set

This is perhaps the best known database to be found in the pattern recognition literature. Fisher's paper is a classic in the field and is referenced frequently to this day. The data set contains 3 classes of 50 instances each, where each class refers to a type of iris plant.

Predicted attribute: class of iris plant.

Attribute Information:

1. sepal length in cm
2. sepal width in cm
3. petal length in cm
4. petal width in cm
5. class: Iris Setosa, Iris Versicolour, Iris Virginica



In [20]:
```
iris
```

Sepal.Length	Sepal.Width	Petal.Length	Petal.Width	Species
5.1	3.5	1.4	0.2	setosa
4.9	3.0	1.4	0.2	setosa
4.7	3.2	1.3	0.2	setosa
4.6	3.1	1.5	0.2	setosa
5.0	3.6	1.4	0.2	setosa
5.4	3.9	1.7	0.4	setosa
4.6	3.4	1.4	0.3	setosa
5.0	3.4	1.5	0.2	setosa
4.4	2.9	1.4	0.2	setosa
4.9	3.1	1.5	0.1	setosa
5.4	3.7	1.5	0.2	setosa
4.8	3.4	1.6	0.2	setosa
4.8	3.0	1.4	0.1	setosa
4.3	3.0	1.1	0.1	setosa
5.8	4.0	1.2	0.2	setosa
5.7	4.4	1.5	0.4	setosa
5.4	3.9	1.3	0.4	setosa
5.1	3.5	1.4	0.3	setosa
5.7	3.8	1.7	0.3	setosa
5.1	3.8	1.5	0.3	setosa
5.4	3.4	1.7	0.2	setosa
5.1	3.7	1.5	0.4	setosa
4.6	3.6	1.0	0.2	setosa
5.1	3.3	1.7	0.5	setosa
4.8	3.4	1.9	0.2	setosa
5.0	3.0	1.6	0.2	setosa
5.0	3.4	1.6	0.4	setosa
5.2	3.5	1.5	0.2	setosa
5.2	3.4	1.4	0.2	setosa
4.7	3.2	1.6	0.2	setosa
...	...	...	...	...
6.9	3.2	5.7	2.3	virginica
5.6	2.8	4.9	2.0	virginica
7.7	2.8	6.7	2.0	virginica
6.3	2.7	4.9	1.8	virginica
6.7	3.3	5.7	2.1	virginica

Sepal.Length	Sepal.Width	Petal.Length	Petal.Width	Species
7.2	3.2	6.0	1.8	virginica
6.2	2.8	4.8	1.8	virginica
6.1	3.0	4.9	1.8	virginica
6.4	2.8	5.6	2.1	virginica
7.2	3.0	5.8	1.6	virginica
7.4	2.8	6.1	1.9	virginica
7.9	3.8	6.4	2.0	virginica
6.4	2.8	5.6	2.2	virginica
6.3	2.8	5.1	1.5	virginica
6.1	2.6	5.6	1.4	virginica
7.7	3.0	6.1	2.3	virginica
6.3	3.4	5.6	2.4	virginica
6.4	3.1	5.5	1.8	virginica
6.0	3.0	4.8	1.8	virginica
6.9	3.1	5.4	2.1	virginica
6.7	3.1	5.6	2.4	virginica
6.9	3.1	5.1	2.3	virginica
5.8	2.7	5.1	1.9	virginica
6.8	3.2	5.9	2.3	virginica
6.7	3.3	5.7	2.5	virginica
6.7	3.0	5.2	2.3	virginica
6.3	2.5	5.0	1.9	virginica
6.5	3.0	5.2	2.0	virginica
6.2	3.4	5.4	2.3	virginica
5.9	3.0	5.1	1.8	virginica

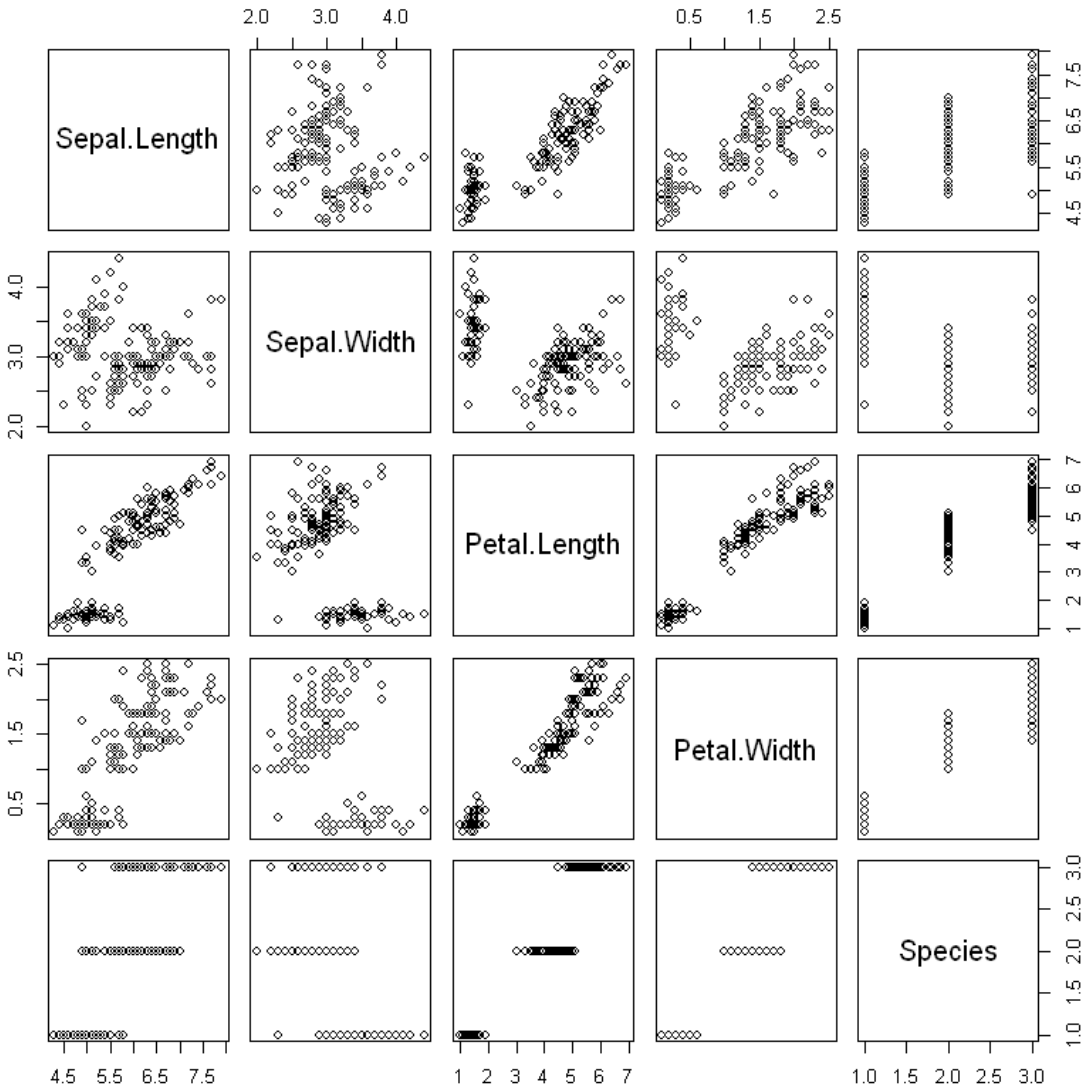
In [21]:
```
summary(iris)
```

```
  Sepal.Length    Sepal.Width     Petal.Length    Petal.Width
 Min.   :4.300   Min.   :2.000   Min.   :1.000   Min.   :0.100
 1st Qu.:5.100   1st Qu.:2.800   1st Qu.:1.600   1st Qu.:0.300
 Median :5.800   Median :3.000   Median :4.350   Median :1.300
 Mean   :5.843   Mean   :3.057   Mean   :3.758   Mean   :1.199
 3rd Qu.:6.400   3rd Qu.:3.300   3rd Qu.:5.100   3rd Qu.:1.800
 Max.   :7.900   Max.   :4.400   Max.   :6.900   Max.   :2.500
       Species
 setosa    :50
 versicolor:50
 virginica :50
```

In [18]:
```
str(iris)
```

```
'data.frame':   150 obs. of  5 variables:
 $ Sepal.Length: num  5.1 4.9 4.7 4.6 5 5.4 4.6 5 4.4 4.9 ...
 $ Sepal.Width : num  3.5 3 3.2 3.1 3.6 3.9 3.4 3.4 2.9 3.1 ...
 $ Petal.Length: num  1.4 1.4 1.3 1.5 1.4 1.7 1.4 1.5 1.4 1.5 ...
 $ Petal.Width : num  0.2 0.2 0.2 0.2 0.2 0.4 0.3 0.2 0.2 0.1 ...
 $ Species     : Factor w/ 3 levels "setosa","versicolor",..: 1 1 1 1 1 1 1 1
1 1 ...
```

In [19]:
```
plot(iris)
```



In [22]:
```
# install.packages("psych")
library(psych)

head(iris)
data <- iris[, 1:4] # Variables numéricas
grupos <- iris[, 5] # Variable factor (grupos)

head(data)
corPlot(data, cex = 1.2, main = "Matriz de correlación")

#https://r-coder.com/grafico-correlacion-r/
```

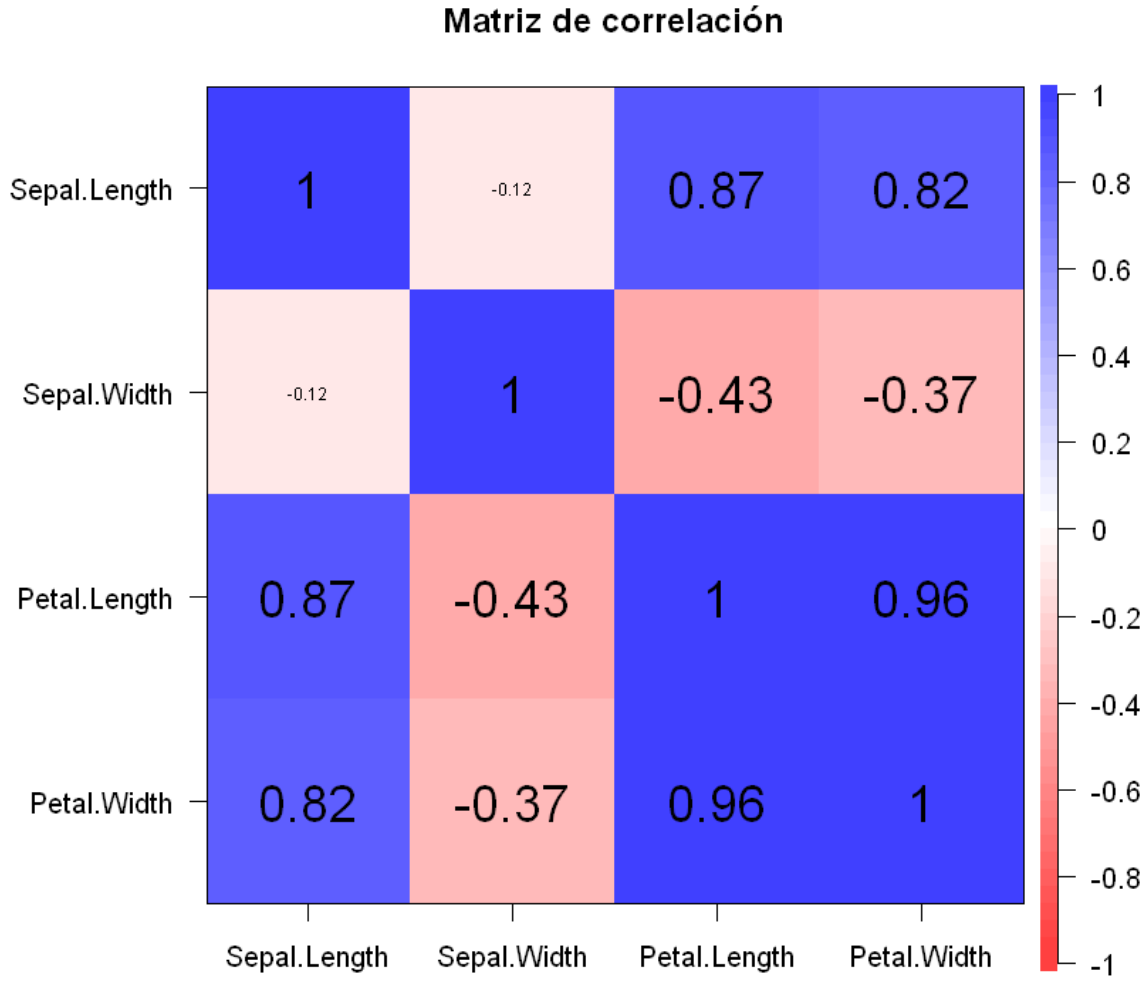
```
Attaching package: 'psych'

The following objects are masked from 'package:ggplot2':

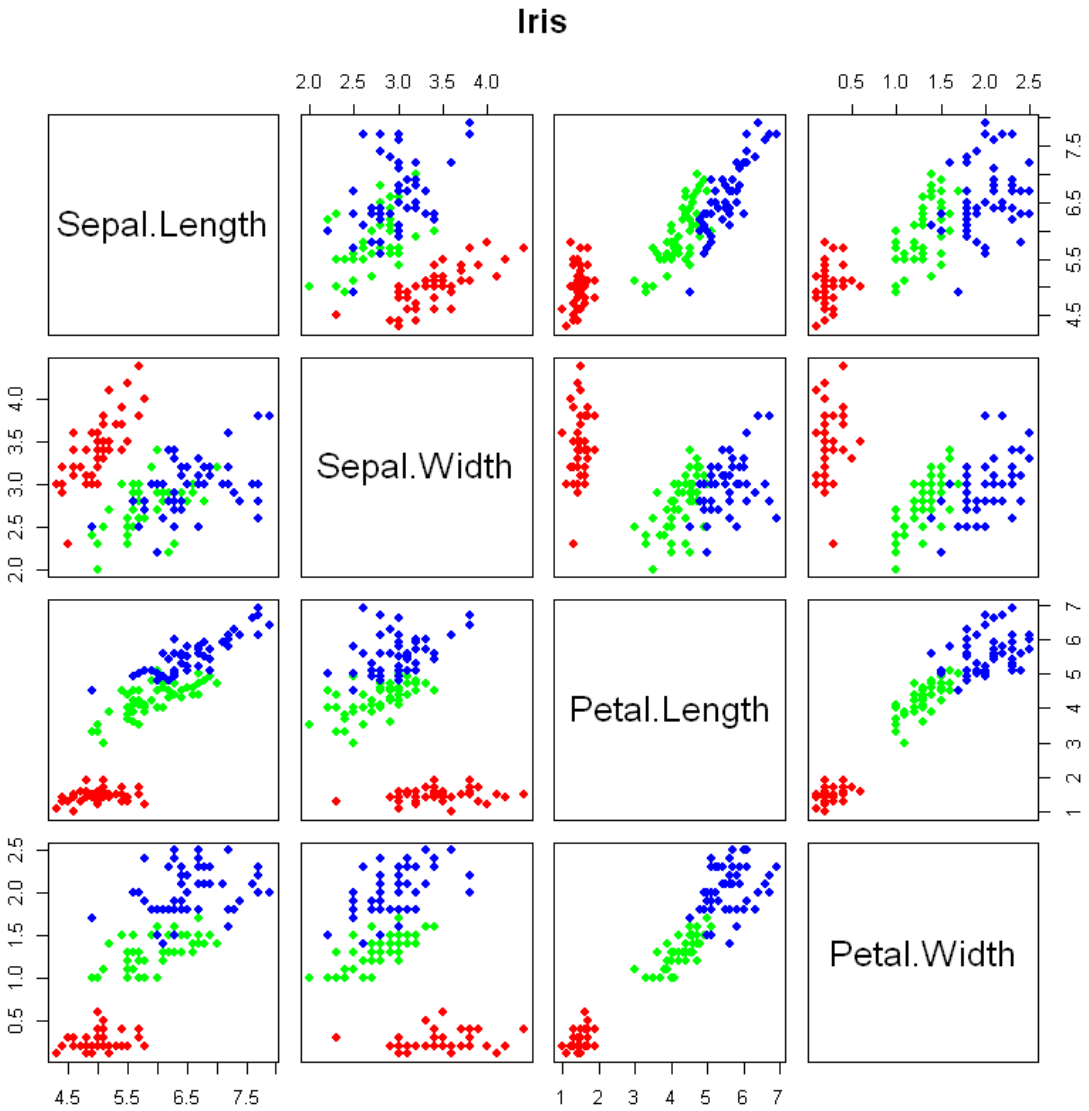
    %+%, alpha
```

Sepal.Length	Sepal.Width	Petal.Length	Petal.Width	Species
5.1	3.5	1.4	0.2	setosa
4.9	3.0	1.4	0.2	setosa
4.7	3.2	1.3	0.2	setosa
4.6	3.1	1.5	0.2	setosa
5.0	3.6	1.4	0.2	setosa
5.4	3.9	1.7	0.4	setosa

Sepal.Length	Sepal.Width	Petal.Length	Petal.Width
5.1	3.5	1.4	0.2
4.9	3.0	1.4	0.2
4.7	3.2	1.3	0.2
4.6	3.1	1.5	0.2
5.0	3.6	1.4	0.2
5.4	3.9	1.7	0.4


Matriz de correlación
Sepal.Length
Sepal.Width
Petal.Length
Petal.Width
1
-0.12
0.87
0.82
-0.12
1
-0.43
-0.37
0.87
-0.43
1
0.96
0.82
-0.37
0.96
1
Sepal.Length
Sepal.Width
Petal.Length
Petal.Width
1
0.8
0.6
0.4
0.2
0
-0.2
-0.4
-0.6
-0.8
-1

In [23]:
```
pairs(data,                       # Data frame de variables
      labels = colnames(data),    # Nombres de las variables
      pch = 21,                   # Símbolo pch
      bg = rainbow(3)[grupos],    # Color de fondo del símbolo (pch 21 a 25)
      col = rainbow(3)[grupos],   # Color de borde del símbolo
      main = "Iris",              # Título del gráfico
      row1attop = TRUE,           # Si FALSE, cambia la dirección de la diagonal
      gap = 1,                    # Distancia entre subplots
      cex.labels = NULL,          # Tamaño del texto de la diagonal
      font.labels = 1)            # Estilo de fuente del texto de la diagonal
```



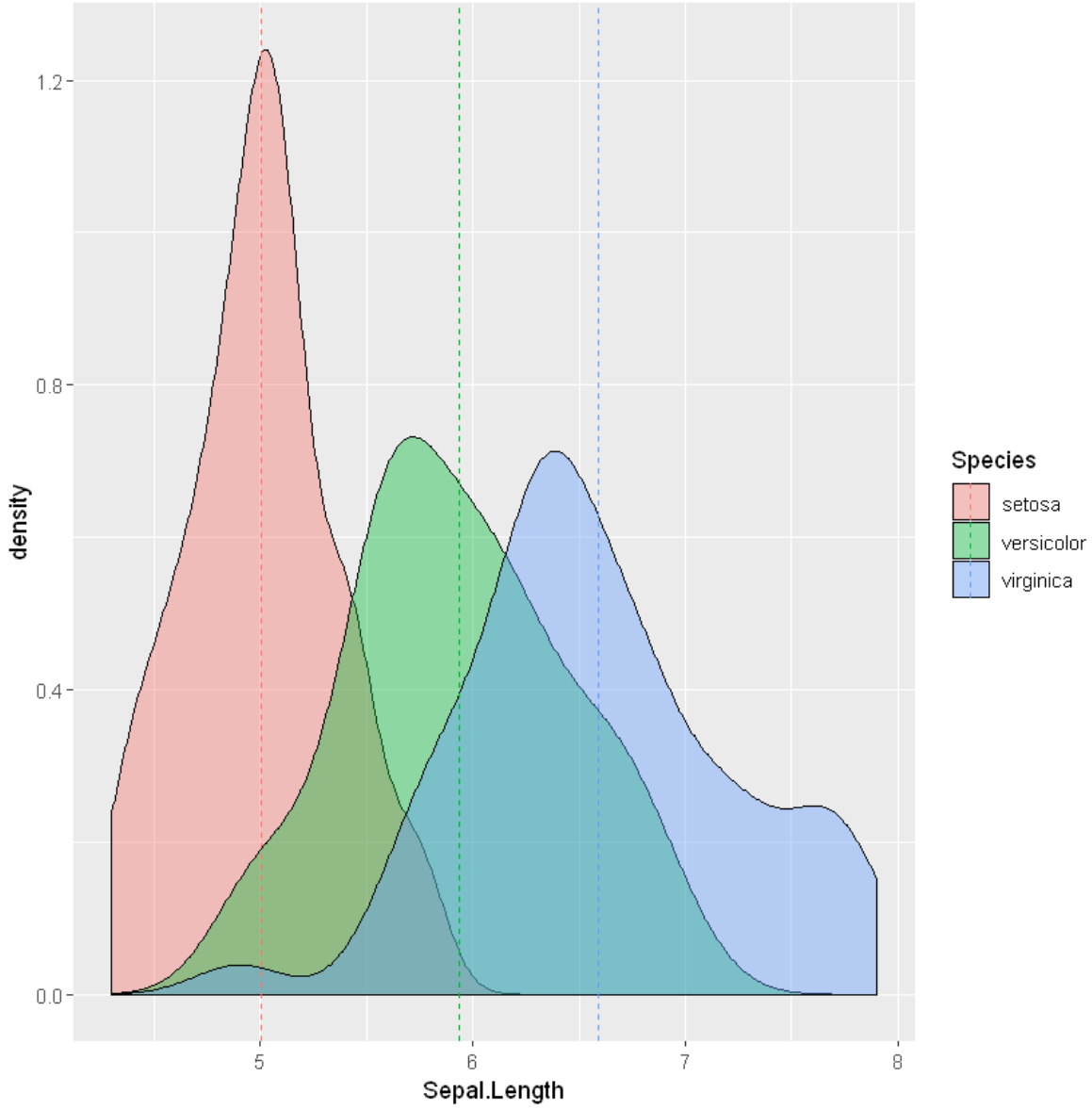
In [24]:

```
library(ggplot2)
library(dplyr)


sl <- iris %>%
group_by(Species)  %>%
summarise(Sepal.Length_mean = mean(Sepal.Length))
#ma


# Change density plot fill colors by groups
#ggplot(iris, aes(x=Sepal.Length, fill=Species)) +
#  geom_density()


# Use semi-transparent fill
p<-ggplot(iris, aes(x=Sepal.Length, fill=Species)) +
  geom_density(alpha=0.4)
#p
# Add mean lines
p+geom_vline(data=sl, aes(xintercept=Sepal.Length_mean, color=Species),
             linetype="dashed")
```



In [ ]:
```
#install.packages("viridis", dep = T)
```

In [ ]:

```
# Vertical box plot by group
# boxplot(Sepal.Length ~ Species, data = iris, col = "white")

# Points
# stripchart(Sepal.Length ~ Species,
#            data = iris,
#             method = "jitter",
#             pch = 19,
#             col = 2:4,
#             vertical = TRUE,
#             add = TRUE)

# ################################################
# # Vertical box plot by group
# boxplot(Sepal.Width  ~ Species, data = iris, col = "white")

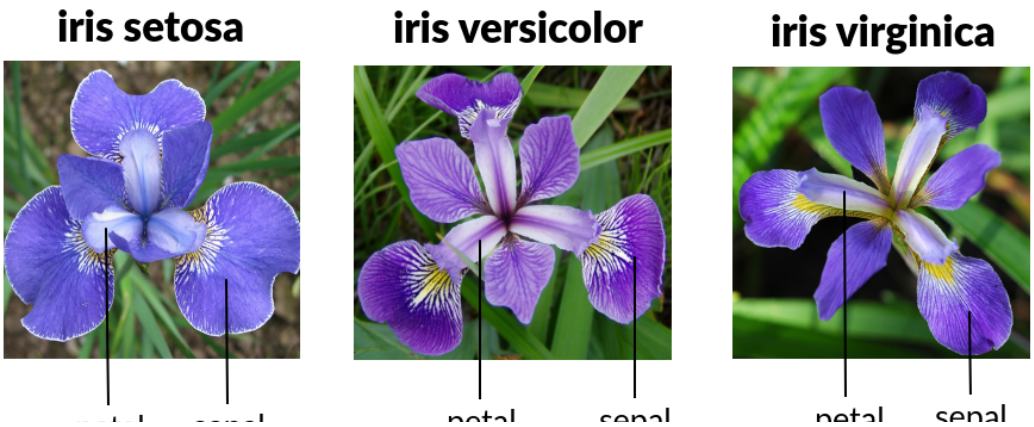
# # Points
# stripchart(Sepal.Width  ~ Species,
#            data = iris,
#             method = "jitter",
#             pch = 19,
#             col = 2:4,
#             vertical = TRUE,
#             add = TRUE)

# ################################################
# # Vertical box plot by group
# boxplot(Petal.Length ~ Species, data = iris, col = "white")

# # Points
# stripchart(Petal.Length ~ Species,
#            data = iris,
#             method = "jitter",
#             pch = 19,
#             col = 2:4,
#             vertical = TRUE,
#             add = TRUE)
# ################################################
# # Vertical box plot by group
# boxplot(Petal.Width  ~ Species, data = iris, col = "white")

# # Points
# stripchart(Petal.Width  ~ Species,
#            data = iris,
#             method = "jitter",
#             pch = 19,
#             col = 2:4,
#             vertical = TRUE,
#             add = TRUE)
```

These 3 species can be considered different in relation to the length or width of the petals?



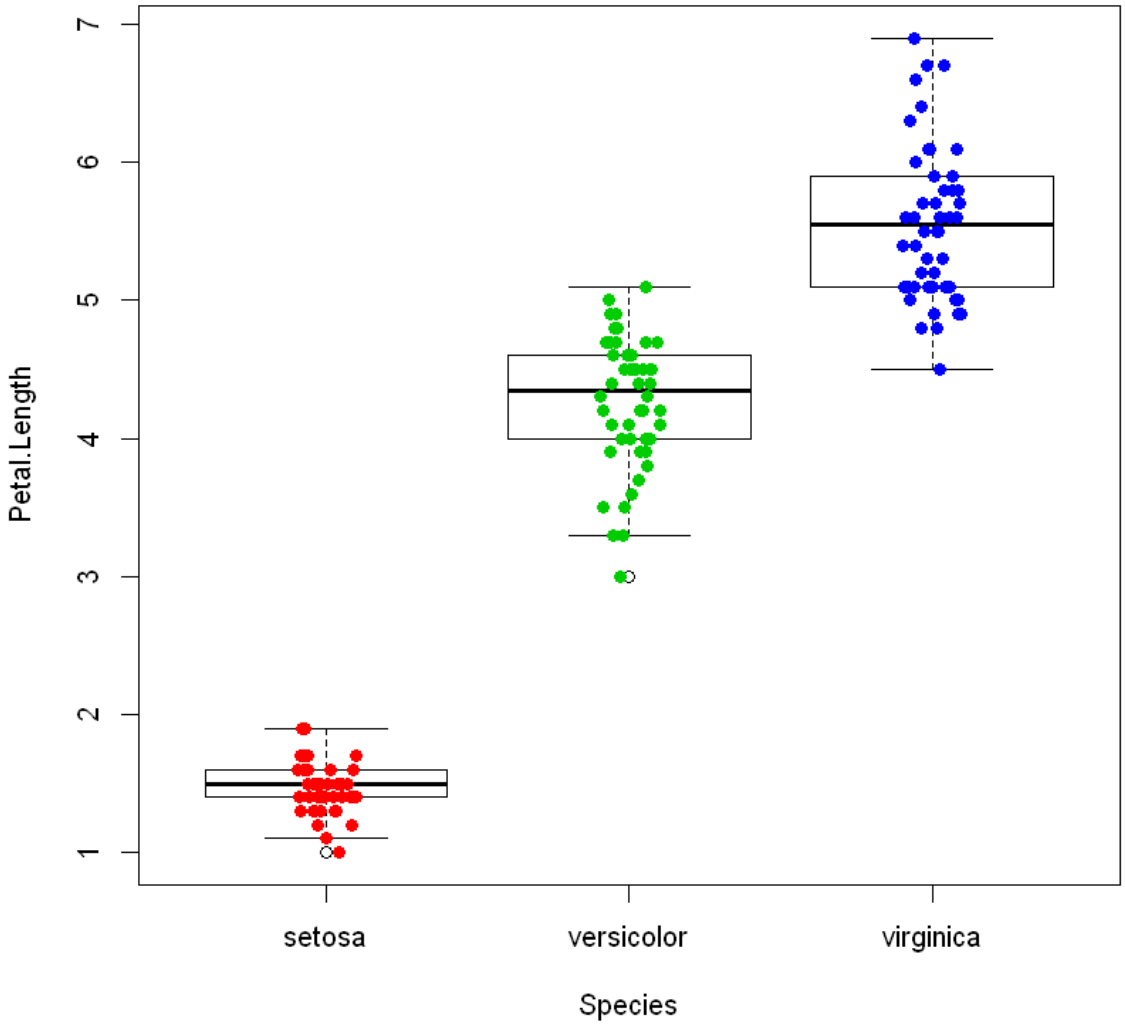
What types of variables do we have?

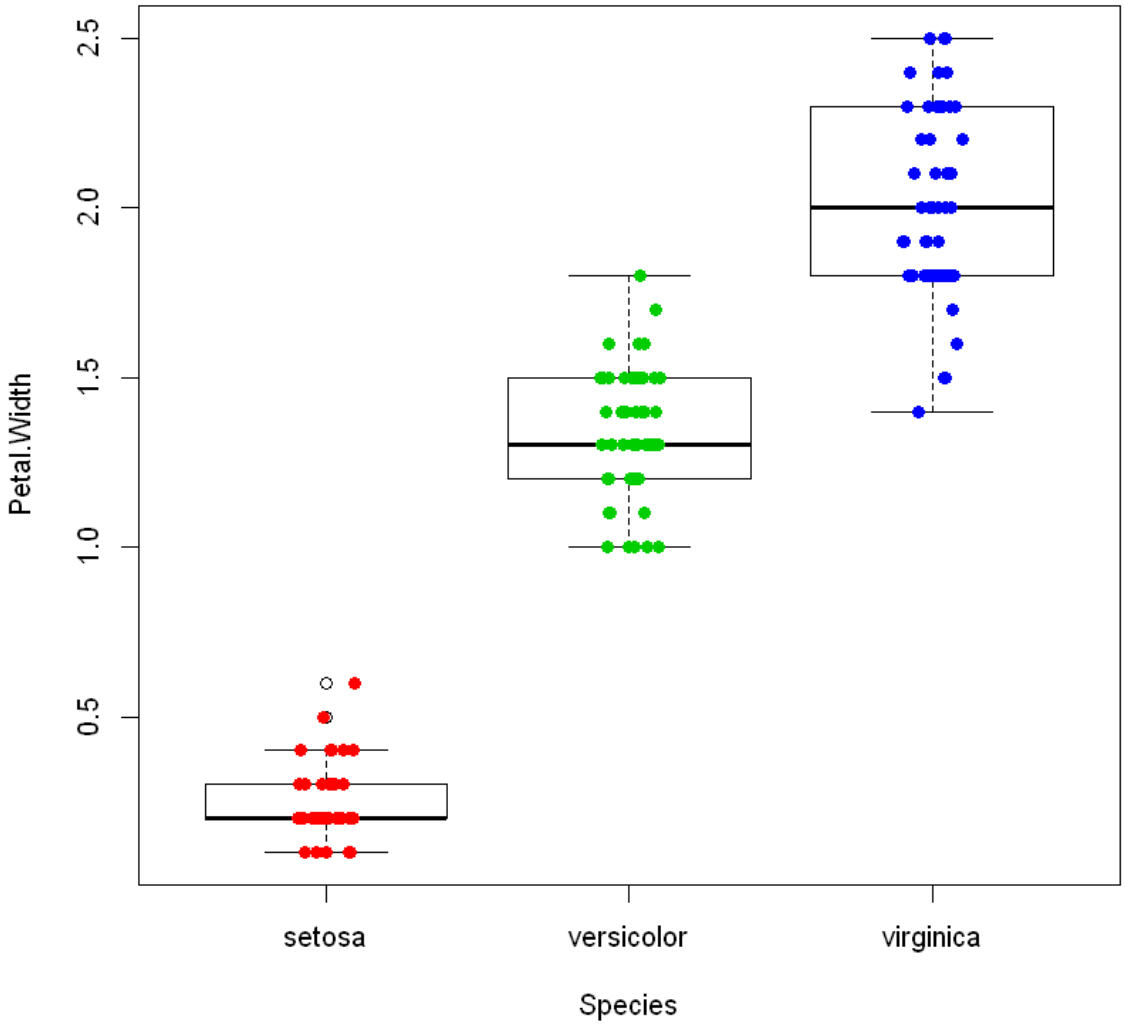
In [25]:

```
##############################################
# Vertical box plot by group
boxplot(Petal.Length ~ Species, data = iris, col = "white")

# Points
stripchart(Petal.Length ~ Species,
           data = iris,
            method = "jitter",
            pch = 19,
            col = 2:4,
            vertical = TRUE,
            add = TRUE)
##############################################
# Vertical box plot by group
boxplot(Petal.Width  ~ Species, data = iris, col = "white")

# Points
stripchart(Petal.Width  ~ Species,
           data = iris,
            method = "jitter",
            pch = 19,
            col = 2:4,
            vertical = TRUE,
            add = TRUE)
```


Petal.Length
1
2
3
4
5
6
7
setosa
versicolor
virginica
Species



ANOVA will report if there are significant differences between the populations of evaluated species

In [28]:
```
results.aov <- aov(iris$Petal.Length ~ iris$Species, data=iris)

summary.aov(results.aov)

#pairwise.t.test(iris$Petal.Length, iris$Species, p.adjust.method = "BH")

TukeyHSD(results.aov)
```

```
             Df Sum Sq Mean Sq F value Pr(>F)
iris$Species   2  437.1  218.55    1180 <2e-16 ***
Residuals    147   27.2    0.19
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

```
  Tukey multiple comparisons of means
    95% family-wise confidence level

Fit: aov(formula = iris$Petal.Length ~ iris$Species, data = iris)

$`iris$Species`
                     diff     lwr     upr p adj
versicolor-setosa    2.798 2.59422 3.00178     0
virginica-setosa     4.090 3.88622 4.29378     0
virginica-versicolor 1.292 1.08822 1.49578     0
```

3.6.2 The Cars data set

In this example, we will take a look at how to implement an ANOVA model to analyse car sales data.

The purpose of this analysis is to determine whether factors such as engine size, horsepower, and fuel efficiency differ across groups of cars based on both vehicle type and country of origin.

A one-way ANOVA is used to determine effects when using just one categorical variable.

A two-way ANOVA is used to determine effects across multiple categories (also known as a factorial ANOVA) and also whether interaction effects are present.

In [29]:
```
### importing the dataset

file <- "Car_sales.csv"

df <- read.csv(file, header = TRUE, sep = ",")

head(df)
```

Manufacturer	Country	Model	Sales_in_thousands	X__year_resale_value	Vehicle_type	Price_in
Acura	Japanese	Integra	16.919	16.360	Passenger	
Acura	Japanese	TL	39.384	19.875	Passenger	
Acura	Japanese	CL	14.114	18.225	Passenger	
Acura	Japanese	RL	8.588	29.725	Passenger	
Audi	German	A4	20.397	22.255	Passenger	
Audi	German	A6	18.780	23.555	Passenger	

In [30]:
```
summary(df)
```

```
    Manufacturer          Country       Model     Sales_in_thousands
Dodge     : 11   American     :70   Neon   :  2   Min.   :  0.11
Ford      : 11   British      : 1   3-Sep  :  1   1st Qu.: 14.11
Chevrolet :  9   German       :24   3000GT :  1   Median : 29.45
Mercedes-B:  9   Japanese     :50   300M   :  1   Mean   : 53.00
Toyota    :  9   Saab         : 2   323i   :  1   3rd Qu.: 67.96
Chrysler  :  7   Sweden       : 6   328i   :  1   Max.   :540.56
(Other)   :101   UnitedKingdom: 4   (Other):150
X__year_resale_value    Vehicle_type Price_in_thousands  Engine_size
Min.   : 5.16        Car      : 41   Min.   : 9.235     Min.   :1.000
1st Qu.:11.26        Passenger:116   1st Qu.:18.017     1st Qu.:2.300
Median :14.18                        Median :22.799     Median :3.000
Mean   :18.07                        Mean   :27.391     Mean   :3.061
3rd Qu.:19.88                        3rd Qu.:31.948     3rd Qu.:3.575
Max.   :67.55                        Max.   :85.500     Max.   :8.000
NA's   :36                           NA's   :2          NA's   :1
  Horsepower      Wheelbase         Width           Length
Min.   : 55.0   Min.   : 92.6   Min.   :62.60   Min.   :149.4
1st Qu.:149.5   1st Qu.:103.0   1st Qu.:68.40   1st Qu.:177.6
Median :177.5   Median :107.0   Median :70.55   Median :187.9
Mean   :185.9   Mean   :107.5   Mean   :71.15   Mean   :187.3
3rd Qu.:215.0   3rd Qu.:112.2   3rd Qu.:73.42   3rd Qu.:196.1
Max.   :450.0   Max.   :138.7   Max.   :79.90   Max.   :224.5
NA's   :1       NA's   :1       NA's   :1       NA's   :1
 Curb_weight    Fuel_capacity   Fuel_efficiency    Latest_Launch
Min.   :1.895   Min.   :10.30   Min.   :15.00   1/24/2011 :  2
1st Qu.:2.971   1st Qu.:15.80   1st Qu.:21.00   1/29/2012 :  2
Median :3.342   Median :17.20   Median :24.00   1/4/2012  :  2
Mean   :3.378   Mean   :17.95   Mean   :23.84   10/20/2009:  2
3rd Qu.:3.800   3rd Qu.:19.57   3rd Qu.:26.00   10/30/2012:  2
Max.   :5.572   Max.   :32.00   Max.   :45.00   10/5/2012 :  2
NA's   :2       NA's   :1       NA's   :3       (Other)   :145
Power_perf_factor
Min.   : 23.28
1st Qu.: 60.41
Median : 72.03
Mean   : 77.04
3rd Qu.: 89.41
Max.   :188.14
NA's   :2
```

In [29]:
```
str(df)
```

```
'data.frame':   157 obs. of  17 variables:
 $ Manufacturer        : Factor w/ 30 levels "Acura","Audi",..: 1 1 1 1 2 2 2
3 3 3 ...
 $ Country             : Factor w/ 7 levels "American","British ",..: 4 4 4 4
3 3 3 3 3 3 ...
 $ Model               : Factor w/ 156 levels "3-Sep","3000GT",..: 80 146 39
121 9 10 11 4 5 8 ...
 $ Sales_in_thousands  : num  16.92 39.38 14.11 8.59 20.4 ...
 $ X__year_resale_value: num  16.4 19.9 18.2 29.7 22.3 ...
 $ Vehicle_type        : Factor w/ 2 levels "Car","Passenger": 2 2 2 2 2 2 2
2 2 2 ...
 $ Price_in_thousands  : num  21.5 28.4 NA 42 24 ...
 $ Engine_size         : num  1.8 3.2 3.2 3.5 1.8 2.8 4.2 2.5 2.8 2.8 ...
 $ Horsepower          : int  140 225 225 210 150 200 310 170 193 193 ...
 $ Wheelbase           : num  101 108 107 115 103 ...
 $ Width               : num  67.3 70.3 70.6 71.4 68.2 76.1 74 68.4 68.5 70.9
...
 $ Length              : num  172 193 192 197 178 ...
 $ Curb_weight         : num  2.64 3.52 3.47 3.85 3 ...
 $ Fuel_capacity       : num  13.2 17.2 17.2 18 16.4 18.5 23.7 16.6 16.6 18.5
...
 $ Fuel_efficiency     : int  28 25 26 22 27 22 21 26 24 25 ...
 $ Latest_Launch       : Factor w/ 130 levels "1/14/2012","1/15/2011",..: 48
94 10 53 21 120 51 92 8 74 ...
 $ Power_perf_factor   : num  58.3 91.4 NA 91.4 62.8 ...
```

Vehicule type vs other variables

In [31]:
```
one.way <- aov(Power_perf_factor ~ Vehicle_type, data = df)
summary(one.way)

one.way <- aov(df$Engine_size ~ Vehicle_type, data = df)
summary(one.way)

one.way <- aov(df$Horsepower ~ Vehicle_type, data = df)
summary(one.way)

one.way <- aov(df$Fuel_capacity ~ Vehicle_type, data = df)
summary(one.way)

one.way <- aov(df$Fuel_efficiency ~ Vehicle_type, data = df)
summary(one.way)
```

```
              Df Sum Sq Mean Sq F value Pr(>F)
Vehicle_type   1      0     0.1       0   0.99
Residuals    153  97352   636.3
2 observations deleted due to missingness

              Df Sum Sq Mean Sq F value Pr(>F)
Vehicle_type   1  11.34  11.338   11.06 0.0011 **
Residuals    154 157.81   1.025
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
1 observation deleted due to missingness

              Df Sum Sq Mean Sq F value Pr(>F)
Vehicle_type   1     11      11   0.003  0.954
Residuals    154 498303    3236
1 observation deleted due to missingness

              Df Sum Sq Mean Sq F value Pr(>F)
Vehicle_type   1  836.4   836.4   85.49 <2e-16 ***
Residuals    154 1506.6     9.8
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
1 observation deleted due to missingness

              Df Sum Sq Mean Sq F value   Pr(>F)
Vehicle_type   1    928   928.0    75.1 6.21e-15 ***
Residuals    152   1878    12.4
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
3 observations deleted due to missingness
```

We can see that for vehicle type (which is specified as either Passenger or Car) in the dataset, the results are not significant for horsepower or power performance factor, i.e. the means of these variables do not show a statistically significant difference depending on whether the vehicle is classified as Passenger or Car.

Let’s now take a look at whether the means of these variables differ across country.

Country vs other variables

In [32]:
```
one.way <- aov(Power_perf_factor ~ df$Country, data = df)
summary(one.way)

one.way <- aov(df$Engine_size ~ df$Country, data = df)
summary(one.way)

one.way <- aov(df$Horsepower ~ df$Country, data = df)
summary(one.way)

one.way <- aov(df$Fuel_capacity ~ df$Country, data = df)
summary(one.way)

one.way <- aov(df$Fuel_efficiency ~ df$Country, data = df)
summary(one.way)
```

```
             Df Sum Sq Mean Sq F value Pr(>F)
df$Country    6   5142     857   1.376  0.228
Residuals   148  92210     623
2 observations deleted due to missingness

             Df Sum Sq Mean Sq F value  Pr(>F)
df$Country    6  17.88   2.981   2.936 0.00978 **
Residuals   149 151.27   1.015
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
1 observation deleted due to missingness

             Df Sum Sq Mean Sq F value Pr(>F)
df$Country    6  16659    2776   0.859  0.527
Residuals   149 481655    3233
1 observation deleted due to missingness

             Df Sum Sq Mean Sq F value Pr(>F)
df$Country    6   50.7   8.448   0.549   0.77
Residuals   149 2292.3  15.384
1 observation deleted due to missingness

             Df Sum Sq Mean Sq F value Pr(>F)
df$Country    6   24.3   4.043   0.214  0.972
Residuals   147 2782.0  18.925
3 observations deleted due to missingness
```

We can see that the p-values for the F-statistic come back as insignificant for all but engine size, meaning that there are statistically significant differences across the mean engine size depending on the country of manufacturer.

It is important to note that one limitation of using ANOVA is that the test will only indicate whether the mean differences across groups are statistically significant - they do not particularly reveal which group is responsible for the differences in means.

For instance, we may observe that mean engine size is indicated to differ across countries — but ANOVA itself cannot tell us which country shows a larger mean engine size across manufacturers as compared to vehicles from other countries.

Two-way ANOVA

Moreover, a limitation of one-way ANOVA is that it cannot test for the effects of interactions. For instance, a vehicle type and manufacturing country may not affect the horsepower of a vehicle individually, but what about when the two factors are combined?

In this regard, a two-way ANOVA is used to 1) test the effect of multiple factors on the dependent variable and 2) the effect of combining these factors, i.e. an interaction effect.

In [33]:
```
model_1 = lm(Engine_size ~ Vehicle_type + Country + Vehicle_type*Country, dat
model_2 = lm(Fuel_efficiency ~ Vehicle_type + Country + Vehicle_type*Country,
model_3 = lm(Fuel_capacity ~ Vehicle_type + Country + Vehicle_type*Country, d
model_4 = lm(Sales_in_thousands ~ Vehicle_type + Country + Vehicle_type*Count
```

A two-way ANOVA test is run on each of the four models:

In [ ]:
```
#install.packages("car")
```

In [34]:
```
library(car)
Anova(model_1, type = "II")
Anova(model_2, type = "II")
Anova(model_3, type = "II")
Anova(model_4, type = "II")
```

```
Loading required package: carData

Attaching package: 'car'

The following object is masked from 'package:psych':

    logit

The following object is masked from 'package:dplyr':

    recode

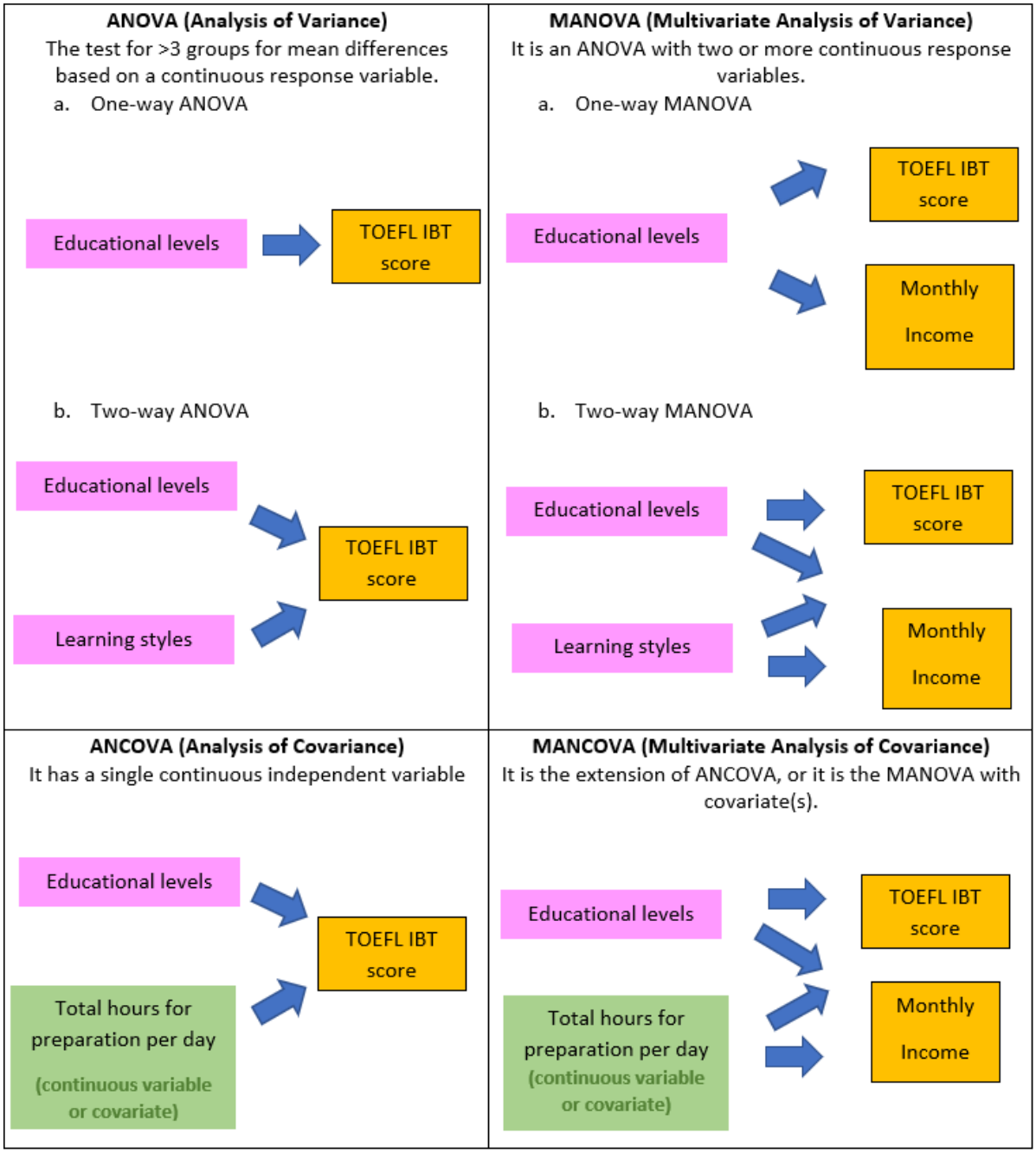
Note: model has aliased coefficients
      sums of squares computed by model comparison
```

	Sum Sq	Df	F value	Pr(>F)
Vehicle_type	10.1102892	1	10.4307950	0.001533253
Country	16.6548366	6	2.8638018	0.011489092

We can see that for model 4, the interaction between vehicle type and country is statistically significant.

This indicates that the sales value is significantly influenced by both vehicle type and the country of the manufacturer. Moreover, the two variables also interact with each other.

3.7 Examples ANCOVA



Now imagine you wanted to do the above analysis but while controlling for other features of the flower's size.

After all, it might be that the species does not influence petal.length specifically, but more generally, the species influences the overall size of the plant.

So the question is: **Controlling for other plant size measures, does species still influence the length of the petal?**

In our analysis, the other metric that represents plant size will be the variable Sepal.Length, which is also available in the iris dataset. So, we specify our extended model just by adding this new covariate.

In [35]:
```
library(car)

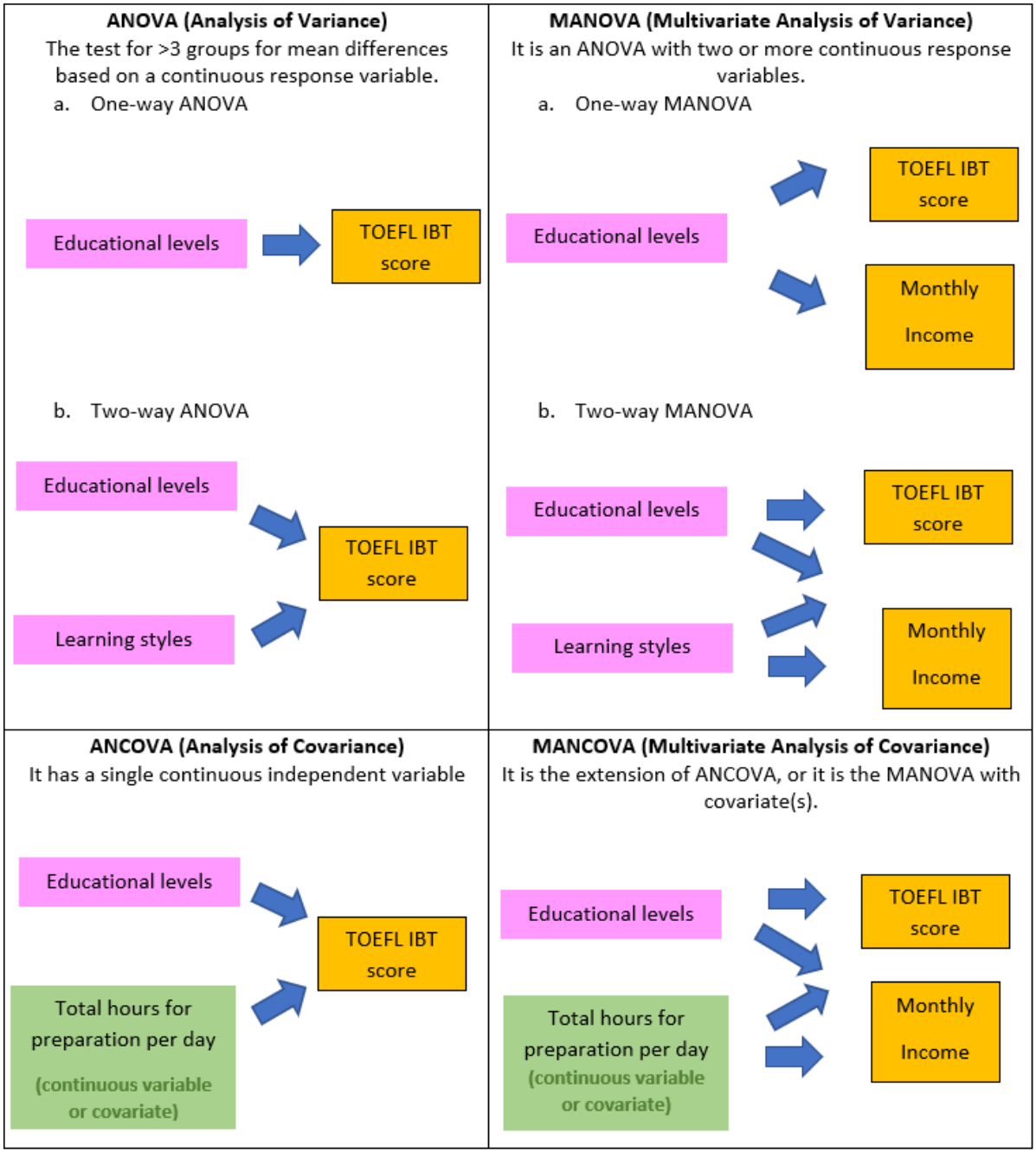
fit2=aov(Petal.Length ~ Species+Sepal.Length, iris)

Anova(fit2, type="III")
```

	Sum Sq	Df	F value	Pr(>F)
(Intercept)	4.369099	1	54.72081	1.005180e-11
Species	99.802009	2	624.98543	2.641099e-72
Sepal.Length	15.565454	1	194.94962	1.121002e-28
Residuals	11.657146	146	NA	NA

As you can see in our row Species, column Pr(>F), which is the p-value, species still has a significant impact on the length of the petal, even when controlling for the length of the sepal. This could imply that the flowers really have different proportions and aren't simply bigger or smaller because of the species.

3.8 Examples MANOVA



In [ ]:

In [36]:
```
library(dplyr)
library(tidyverse)

iris2 <- iris %>%
  select(Sepal.Length, Petal.Length, Species) %>%
  add_column(id = 1:nrow(iris), .before = 1)
head(iris2)
```

```
Registered S3 method overwritten by 'rvest':
  method            from
  read_xml.response xml2
-- Attaching packages --------------------------------------- tidyverse 1.
2.1 --
v tibble  2.1.1     v purrr   0.3.2
v tidyr   0.8.3     v stringr 1.4.0
v readr   1.3.1     v forcats 0.4.0
-- Conflicts ------------------------------------------ tidyverse_conflicts
() --
x psych::%+%()     masks ggplot2::%+%()
x psych::alpha()   masks ggplot2::alpha()
x dplyr::filter()  masks stats::filter()
x purrr::flatten() masks jsonlite::flatten()
x dplyr::lag()     masks stats::lag()
x car::recode()    masks dplyr::recode()
x purrr::some()    masks car::some()
```

id	Sepal.Length	Petal.Length	Species

In [37]:
```
library(car)

model <- lm(cbind(Sepal.Length, Petal.Length) ~ Species, iris2)

Anova(model, type="III")
```

```
Type III MANOVA Tests: Pillai test statistic
            Df test stat approx F num Df den Df    Pr(>F)
(Intercept)  1   0.97807   3255.4      2    146 < 2.2e-16 ***
Species      2   0.98850     71.8      4    294 < 2.2e-16 ***
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

In [ ]:

In [ ]:

In [ ]:

In [ ]:

In [ ]:

3.9 Examples MANCOVA



In [1]:
```
install.packages("rlang")
install.packages("jmv")
```

```
  There is a binary version available but the source version is later:
      binary source needs_compilation
rlang 0.4.11  1.0.2              TRUE

  Binaries will be installed
package 'rlang' successfully unpacked and MD5 sums checked

Warning message:
"cannot remove prior installation of package 'rlang'"Warning message in fil
e.copy(savedcopy, lib, recursive = TRUE):
"problem copying C:\ProgramData\Anaconda3\envs\R\Lib\R\library\00LOCK\rlang
\DESCRIPTION to C:\ProgramData\Anaconda3\envs\R\Lib\R\library\rlang\DESCRIP
TION: Permission denied"Warning message in file.copy(savedcopy, lib, recurs
ive = TRUE):
"problem copying C:\ProgramData\Anaconda3\envs\R\Lib\R\library\00LOCK\rlang
\help\aliases.rds to C:\ProgramData\Anaconda3\envs\R\Lib\R\library\rlang\he
lp\aliases.rds: Permission denied"Warning message in file.copy(savedcopy, l
ib, recursive = TRUE):
"problem copying C:\ProgramData\Anaconda3\envs\R\Lib\R\library\00LOCK\rlang
```

In [3]:
```
library(jmv)

mancova(data = iris,
    deps = vars(Sepal.Length, Sepal.Width, Petal.Length, Petal.Width),
    factors = Species)
```

```
Error in library(jmv): there is no package called 'jmv'
Traceback:

1. library(jmv)
```

In [2]:
```
#https://www.rdocumentation.org/packages/jmv/versions/2.3.4/topics/mancova
```

In [38]:
```
summary.aov(manova(cbind(Sepal.Length,Sepal.Width) ~ Species*Petal.Length, ir
```

```
 Response Sepal.Length :
                     Df Sum Sq Mean Sq  F value Pr(>F)
Species               2 63.212 31.6061 279.2076 <2e-16 ***
Petal.Length          1 22.275 22.2745 196.7730 <2e-16 ***
Species:Petal.Length  2  0.381  0.1905   1.6828 0.1895
Residuals           144 16.301  0.1132
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

 Response Sepal.Width :
                     Df  Sum Sq Mean Sq F value    Pr(>F)
Species               2 11.3449  5.6725 56.7086 < 2.2e-16 ***
Petal.Length          1  2.4225  2.4225 24.2184 2.322e-06 ***
Species:Petal.Length  2  0.1354  0.0677  0.6768    0.5098
Residuals           144 14.4041  0.1000
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```